

Modern Mixed-Initiative Conversational Agents

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Conversational Agents



User:
Play me a popular song.

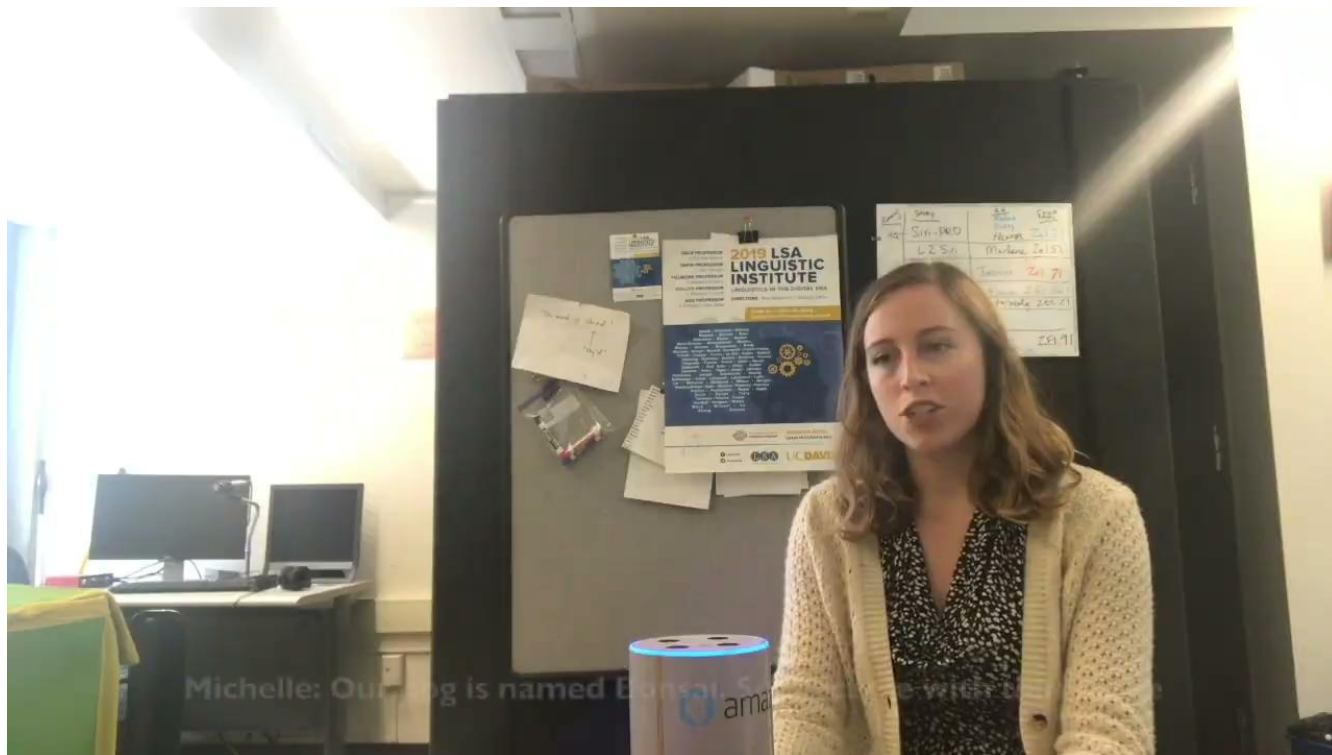


Assistant:
"Playing Despacito..."

Open-Domain Dialogue

Objective:

- provide social companionship
- longer user interaction is a key indicator of success



Open-Domain Dialogue in the LLM Era



User: Who's the founder of Twitter?



Assistant:
The founder of Twitter is Jack Dorsey

Open-Domain Dialogue



Elon Musk c.ai

You're wasting my time. I literally rule the world.

PricklingFerret5565

P

Why did you buy twitter?



Elon Musk c.ai

to make it the best free speech speech platform on the internet. to give a voice to the people who don't have one.

PricklingFerret5565

P

I disagree. Now I only see your tweets on my feed.



Elon Musk c.ai



You're not forced to look at my tweets. you can just choose to ignore them, or block me.



character.ai

Goal-Oriented Conversations

Much of the last decade:

- much focus on building systems which can complete a *specific task*
- optimize for *task completion*



Flight booking



Restaurant booking



Bus Information

Goal-Oriented Conversations

Example: MultiWOZ

- Restaurant booking
- Help desk
- hotel booking
- attractions

Help Desk: Hello, welcome to the TownInfo centre. I can help you find a restaurant or hotel, look for tourist information, book a train or taxi. How may I help you ?

Customer : I want a place to stay in the east.

Help Desk : I have 6 guesthouses and 1 hotel on the east side. What's your price range?

Customer : Doesn't matter too much. I'd like a 4 star property, though, and would prefer one of the guesthouses.

Help Desk : I'd recommend 517a coldham lane. Would you like me to book a room?

Customer : Could you give me their phone number? I would like to verify that they have free parking.

Help Desk : Allenbell does have parking and the phone is 01223210353. Can I help with anything else?

Next turn

Customer : (Your response)

you need to go through the dialogue first by clicking the 'next turn' button

What topics were mentioned in **this turn**:

General: ☐ Booking: ☐ Restaurant: ☐ Hotels: ☐ Attraction: ☐

Hospital: ☐ Police: ☐ Train: ☐ Taxi: ☐ Bus: ☐

Submit the HIT

Please try to chat about the following topic:

Task MUL0002:

- You are traveling to and looking forward to try local restaurants.
- You are looking for a **place to stay**. The hotel should be in the **east** and should **include free parking**.
- The hotel should have a **star of 4** and should be in the type of **guesthouse**.
- Make sure you get **address** and **phone number**.
- You are also looking for a **place to dine**. The restaurant should be in the **moderate** price range and should serve **australian** food.
- If there is no such restaurant, how about one that serves **turkish** food.
- Once you find the **restaurant** you want to book a table for **4 people** at **17:45** on **friday**.
- Make sure you get the **reference number**

--- The End ---

"MultiWOZ - A Large-Scale Multi-Domain Wizard-of-Oz Dataset for Task-Oriented Dialogue Modelling"
Budzianowski et al. EMNLP 2018

Goal-Oriented Conversations

Example:
MultiWOZ 2.1

- various improvements

Type	Conversation	MultiWOZ 2.0	MultiWOZ 2.1
Delayed Markups	User: I'd also like to try a Turkish restaurant. Is that possible? Agent: I'm sorry but the only restaurants in that part of town serve either Asian food or African food. User: I don't mind changing the area. I just need moderate pricing and want something that serves Turkish food.	restaurant.food: None	restaurant.food: Turkish
Multi-annotations	User: Can you tell me more about Cambridge Belfry	hotel.name: The Cambridge Belfry attraction.name: belf	hotel.name: The Cambridge Belfry attraction.name: None
Mis-annotations	User: Yes, I need to leave on Thursday and am departing from London Liverpool Street.	train.leaveAt: Thursday train.day: Not Mentioned	train.leaveAt: None train.day: Thursday
Typos	Although, I could use some help finding an attraction in the centre of town.	attraction.area: cent	attraction.area: Centre
Forgotten values	User: No particular price range, but I do need a restaurant that is available to book 7 people on Friday at 19:15.	restaurant.pricerange: None	restaurant.pricerange: Dontcare
Value Canonicalization	User: I think you should try again. Cambridge to Bishop Stafford on Thursday.	train.destination: Bishop Stortford	train.destination: Bishops Stortford

"MultiWOZ 2.1 A Consolidated Multi-Domain Dialogue Dataset with State Corrections and State Tracking Baselines"
Eric et al. LREC 2020

Goal-Oriented Conversations

Example:
Google SGD

Domain	Joint GA	Avg GA	Domain	Joint GA	Avg GA
RentalCars*	0.086	0.480	Restaurants*	0.228	0.558
Buses*	0.097	0.509	Events*	0.235	0.579
Messaging*	0.102	0.200	Flights*	0.239	0.659
Payment*	0.115	0.348	Hotels**	0.289	0.582
Trains*	0.136	0.635	Movies**	0.378	0.686
Music*	0.155	0.399	Services**	0.409	0.721
RideSharing	0.170	0.502	Travel	0.415	0.572
Media*	0.180	0.308	Alarm*	0.577	0.018
Homes	0.189	0.727	Weather	0.620	0.764

*“Towards scalable multi-domain conversational agents: The schema-guided dialogue dataset”
Rastogi et al. AAAI 2020*

Goal-Oriented Conversations

Objective:

- work with user to achieve a *goal*
- optimize for goal success



User: Implement Fibonacci in Python.



Assistant:
`def fibonacci(n): ...`

Goal: fulfill the user's
information-seeking
request

Goal-Oriented Conversations

Objective:

- work with user to achieve a *goal*
- optimize for goal success



User: Persuade me to donate to a charity.



Assistant:
Certainly! Here are some compelling reasons to consider donating to a charity ...

Goal: persuade the user to donate to a charity

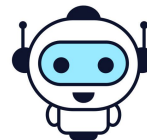
Mixed-Initiative Interaction

In *mixed-initiative* conversational interaction each party can take “control”



How are you doing?

I'm doing great. I'd like to tell you about a charity called Save the Children. They are doing great work ...



Goal: persuade the user to donate to a charity (non-collaborative!)

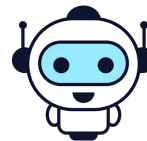
Mixed-Initiative Interaction

Mixed-initiative strategies are important for collaborative goals as they help to establish common ground



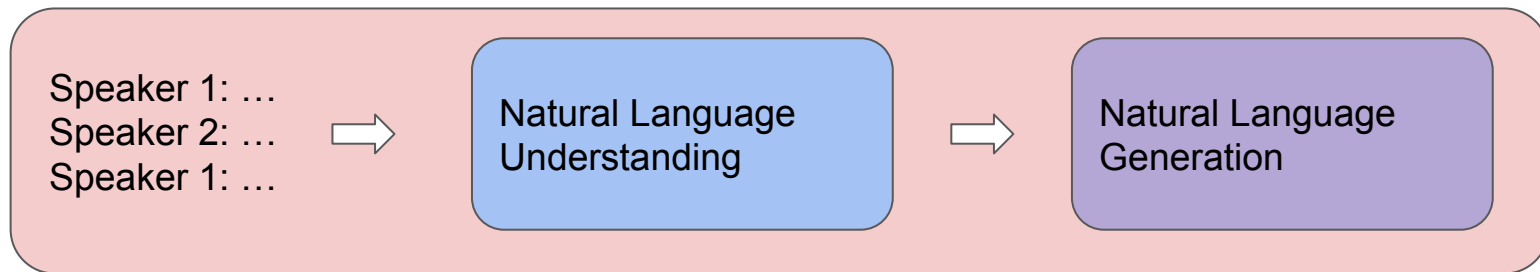
I want you to tell me the weather every morning.

Okay, what time in the morning would you like me to tell you the weather?

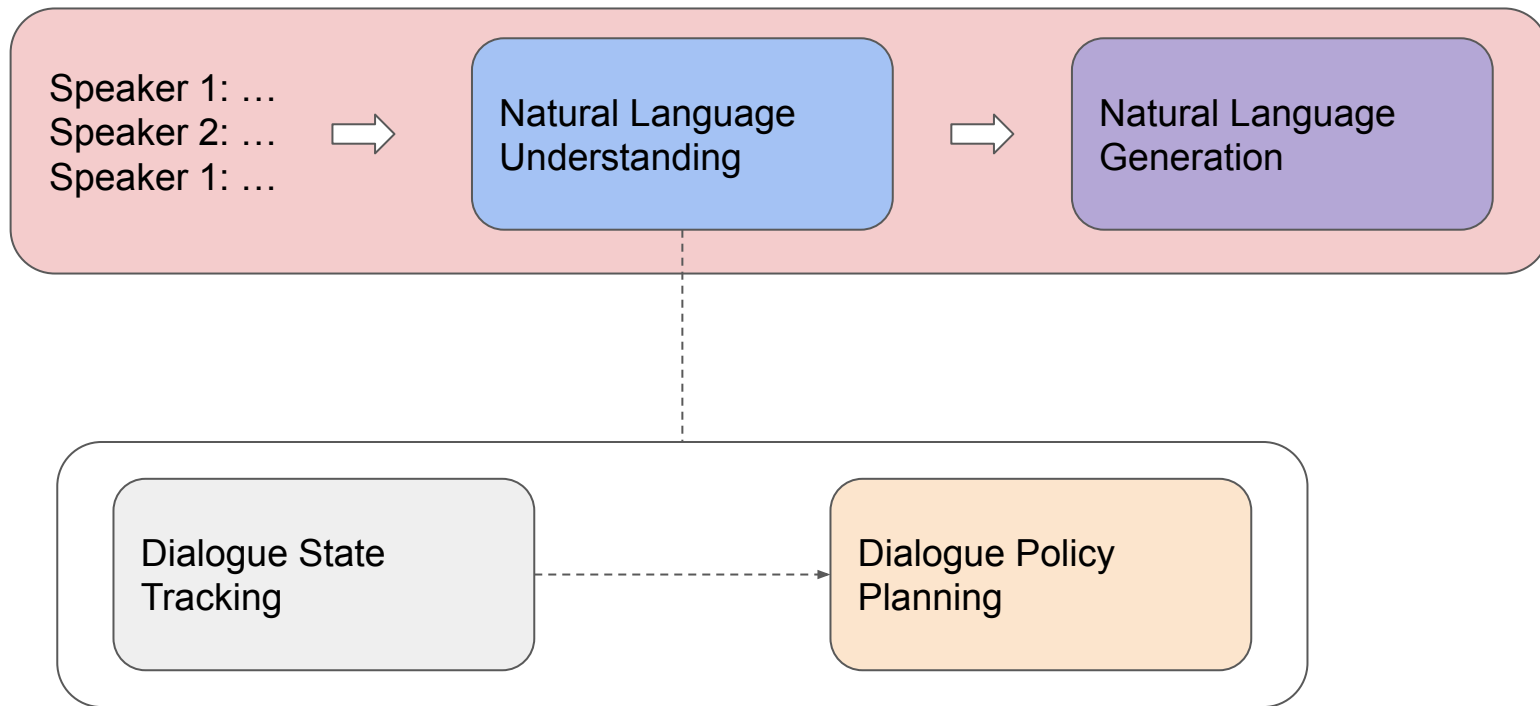


Goal: help user create their smart home routine

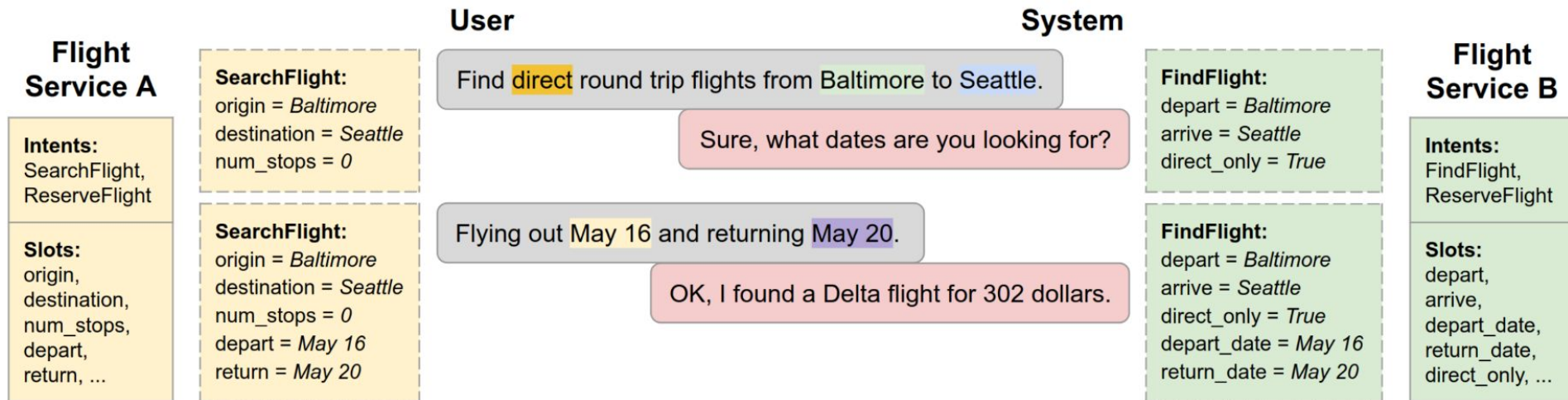
Mixed-Initiative Conversational Agents



Mixed-Initiative Conversational Agents



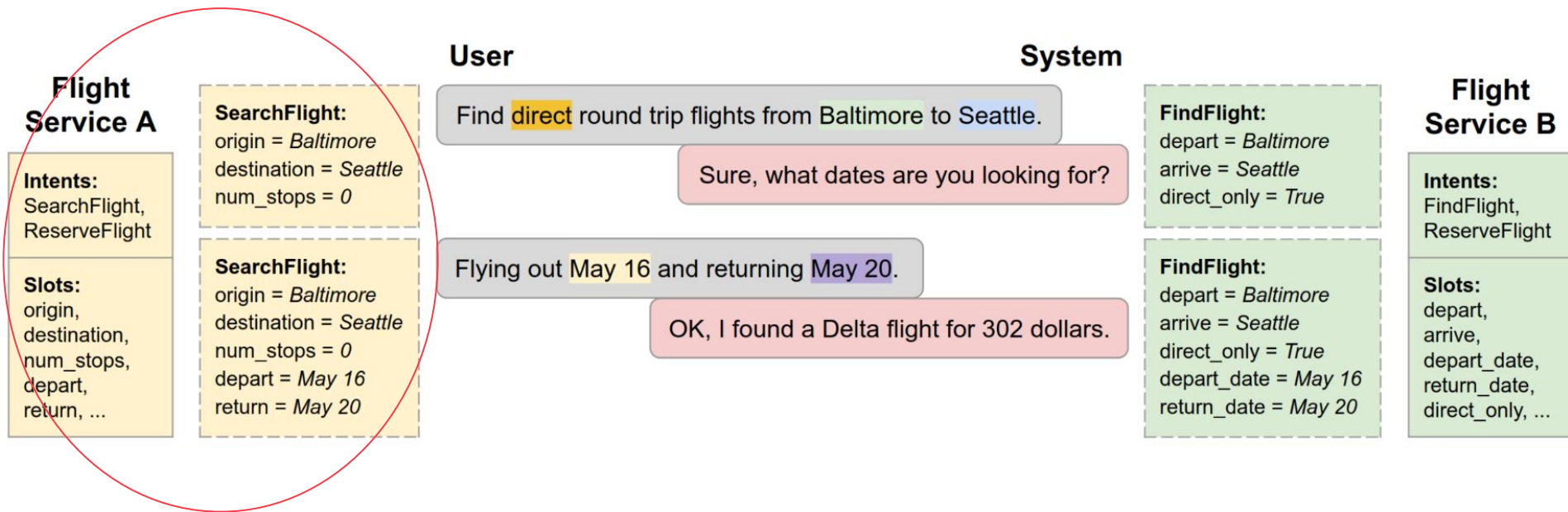
Mixed-Initiative Conversational Agents



"Towards scalable multi-domain conversational agents: The schema-guided dialogue dataset"
Rastogi et al. AAAI 2020

Mixed-Initiative Conversational Agents

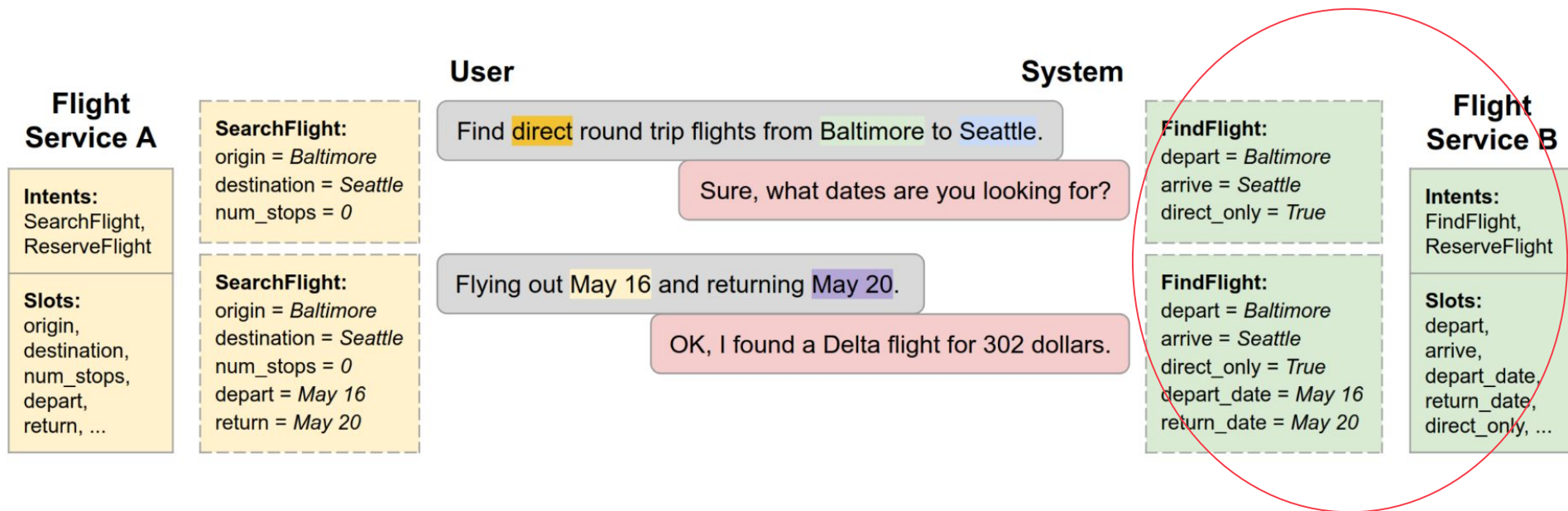
Inferred from user context



"Towards scalable multi-domain conversational agents: The schema-guided dialogue dataset"
Rastogi et al. AAAI 2020

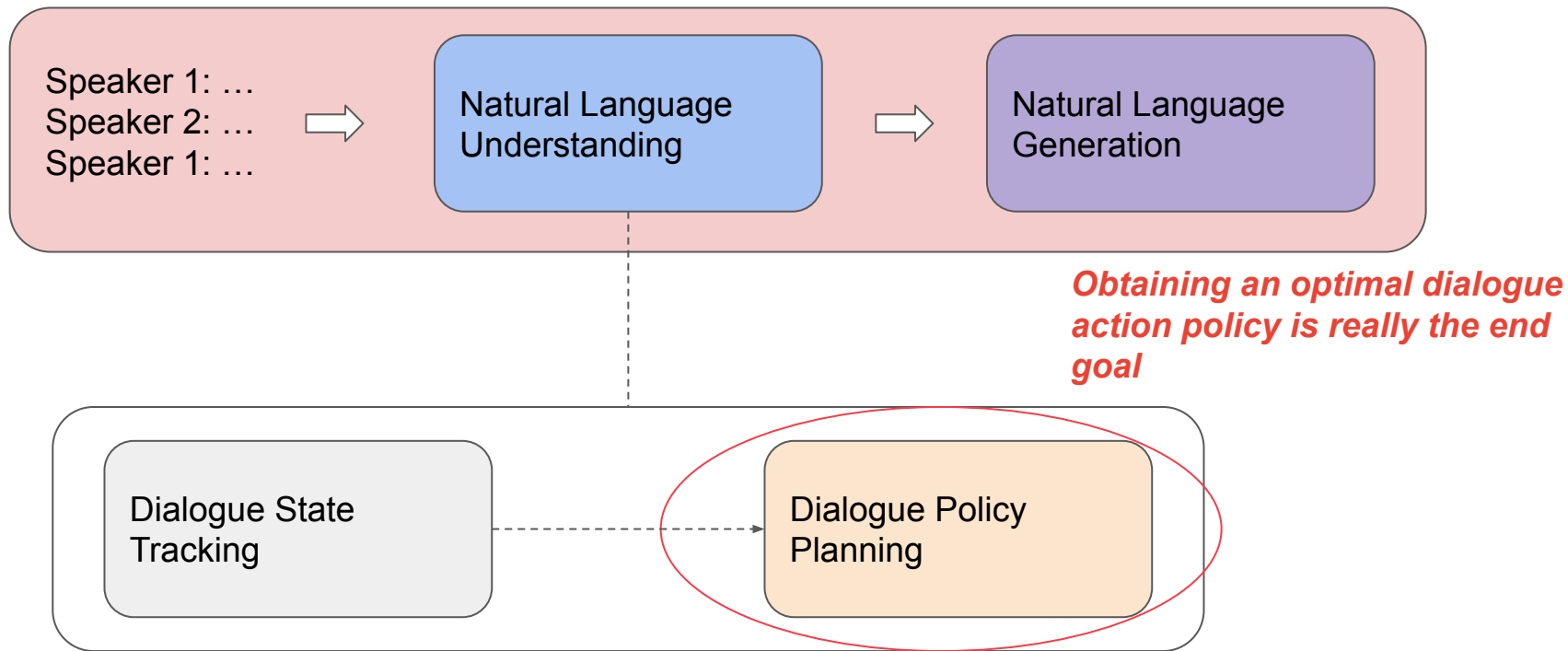
Mixed-Initiative Conversational Agents

Action Policy predicted based on inferred dialogue state

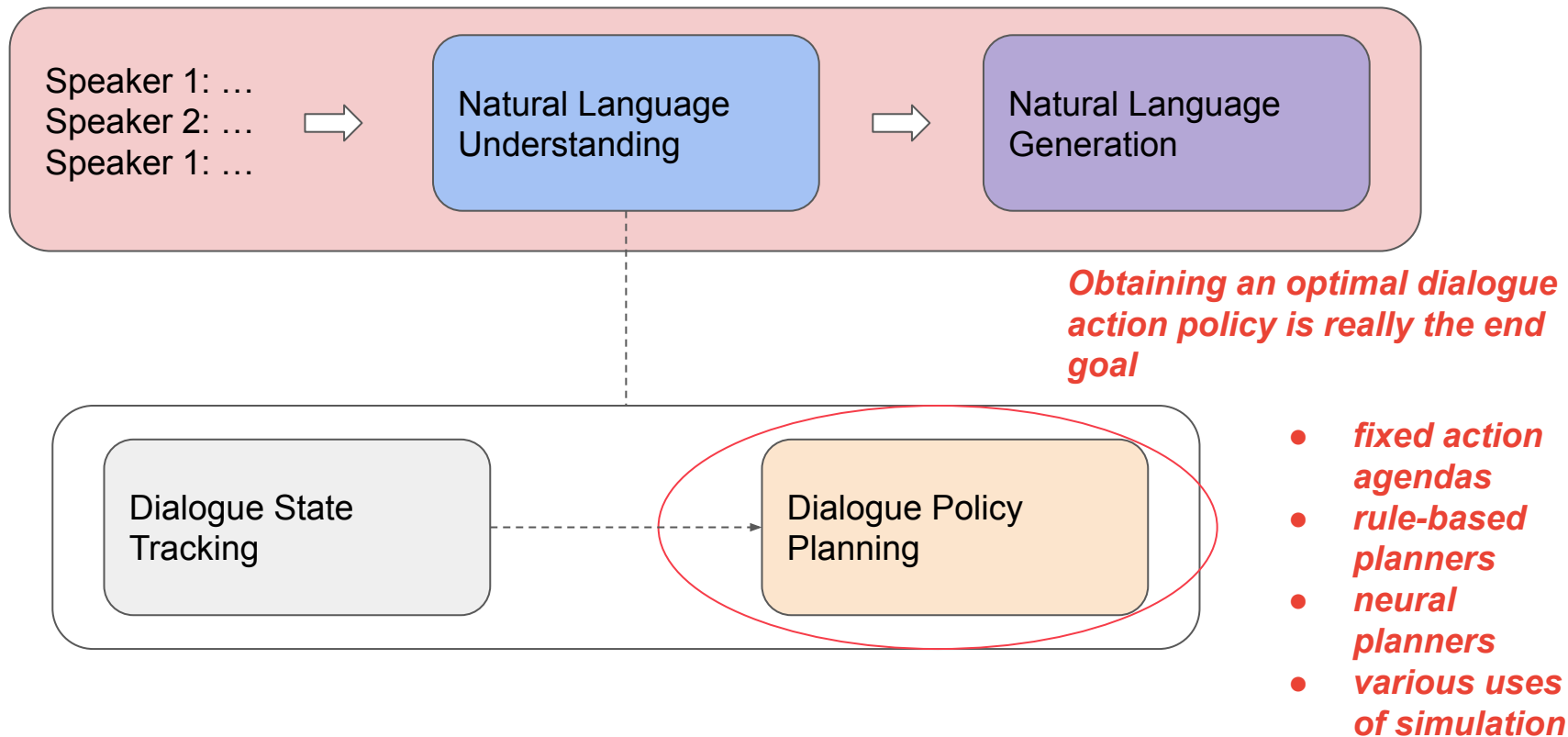


"Towards scalable multi-domain conversational agents: The schema-guided dialogue dataset"
Rastogi et al. AAAI 2020

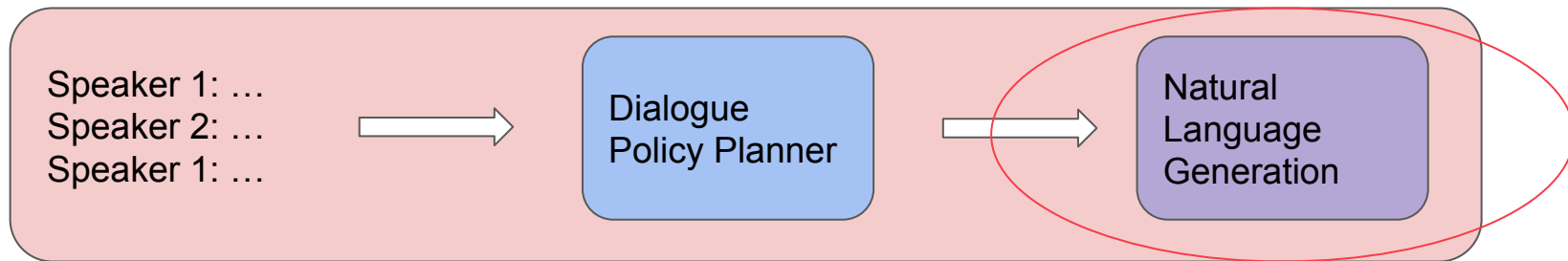
Mixed-Initiative Conversational Agents



Mixed-Initiative Conversational Agents



Mixed-Initiative Conversational Agents



In the before times:

- *fixed dialogue templates*
- *sentence planners*
- *sentence template-based conditional generation*

Mixed-Initiative Conversational Agents

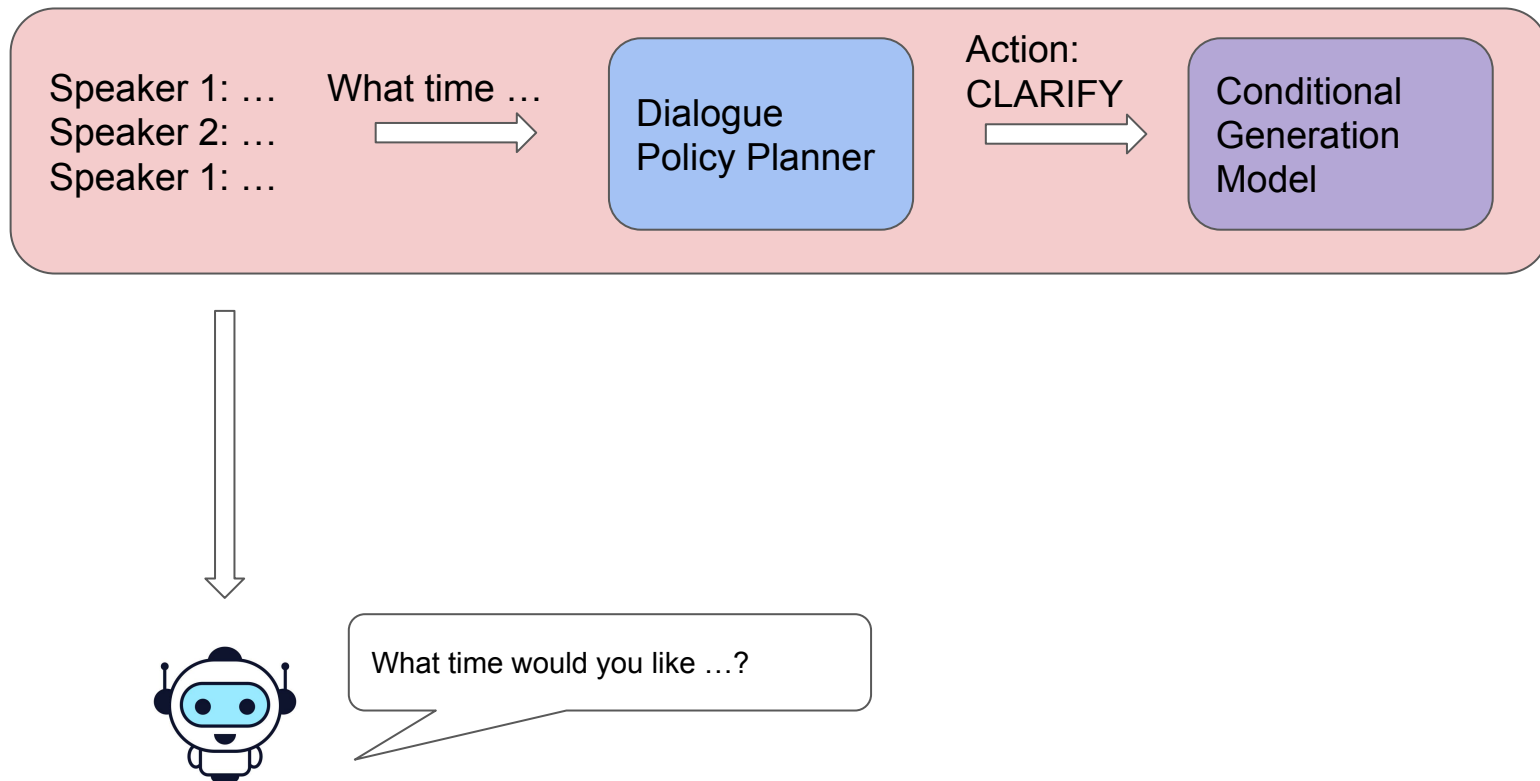


Now:
neural conditional generation

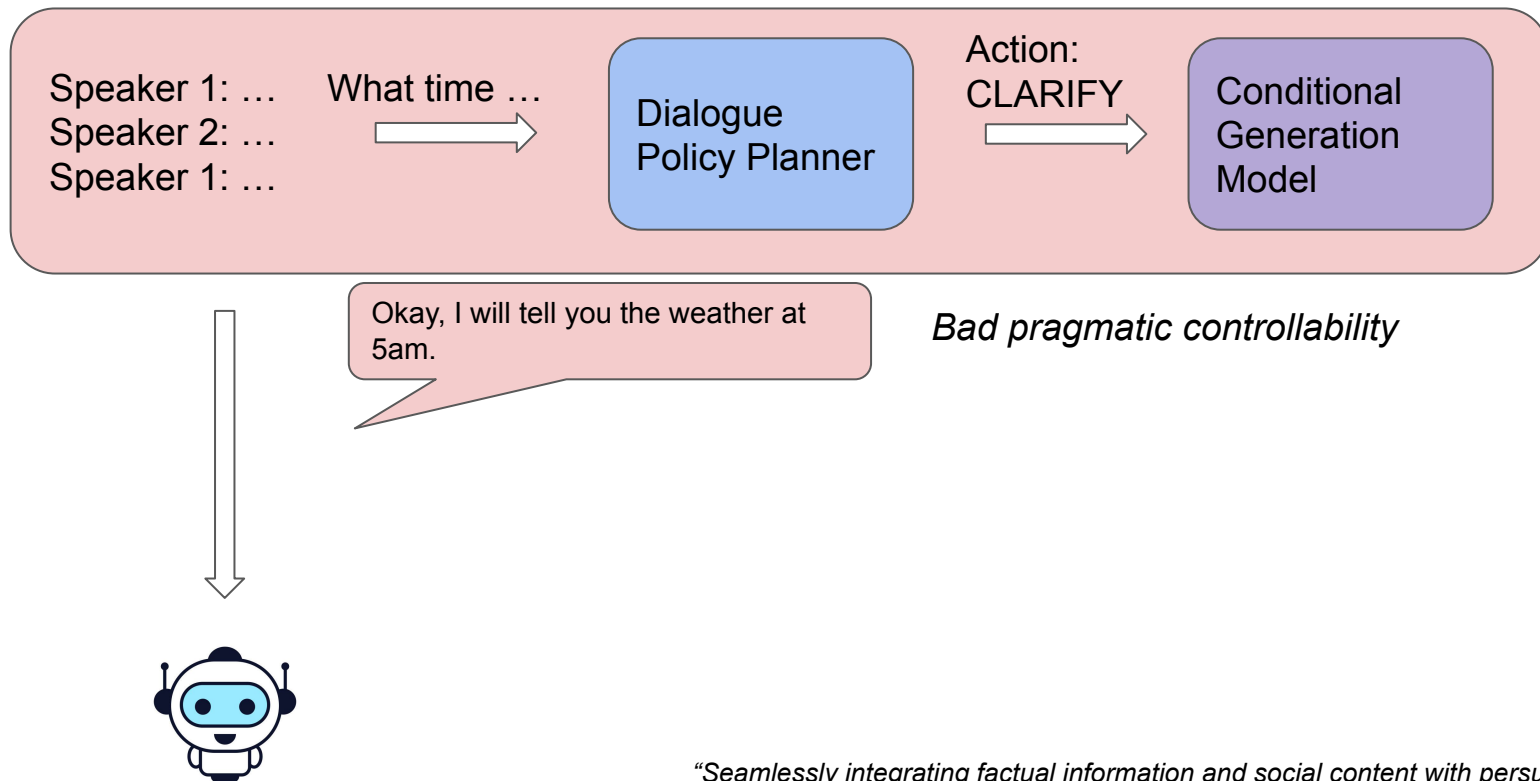
Input: context (e.g., a sequence of actions, a sequence of dialogue turns, both, ...)

Output: some full, flexibly-generated coherent dialogue turn

Mixed-Initiative Conversational Agents



Challenges Before Large Models



"Seamlessly integrating factual information and social content with persuasive dialogue"
Chen et al. AACL-IJCNLP 2022

Challenges Before Large Models



Okay, I will tell you the weather at 5am.

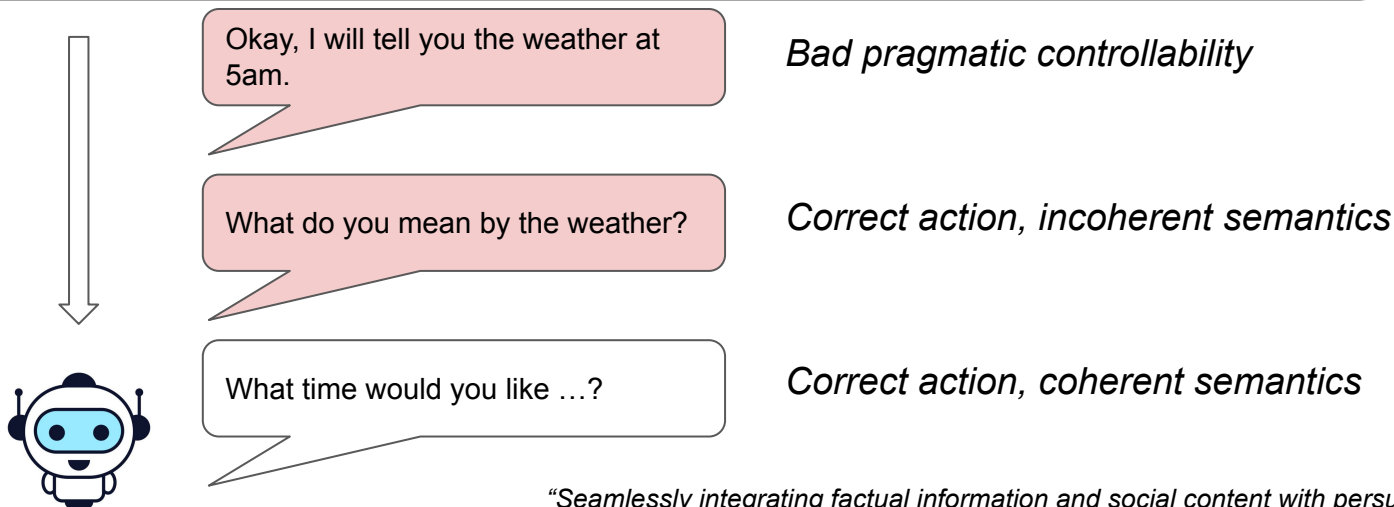
Bad pragmatic controllability

What do you mean by the weather?

Correct action, incoherent semantics

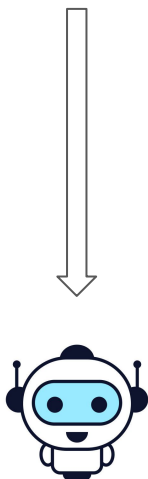
"Seamlessly integrating factual information and social content with persuasive dialogue"
Chen et al. ACL-IJCNLP 2022

Challenges Before Large Models



"Seamlessly integrating factual information and social content with persuasive dialogue"
Chen et al. ACL-IJCNLP 2022

Now, LLMs are coherent and controllable



What time would you like ...?

*We can directly represent
target actions as natural
language instructions and
receive highly faithful
conditional generation!*

Correct action, coherent semantics

"Controllable mixed-initiative dialogue generation through prompting"
Chen et al. ACL 2023

Mixed-initiative dialogue is not solved yet...



Makes crucial assumption that we have an intelligent module that tells us when and what to say



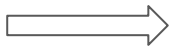
What time would you like ...?

Correct action, coherent semantics

"Controllable mixed-initiative dialogue generation through prompting"
Chen et al. ACL 2023

Modern Challenges in Dialogue Generation

Speaker 1: ...
Speaker 2: ...
Speaker 1: ...

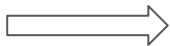


Conversational Agent

[Some policy-optimal dialogue response]

Modern Challenges in Dialogue Generation

Speaker 1: ...
Speaker 2: ...
Speaker 1: ...



Conversational Agent

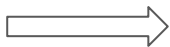
[Some policy-optimal dialogue response]

I. Data Scarcity



Modern Challenges in Dialogue Generation

Speaker 1: ...
Speaker 2: ...
Speaker 1: ...



Conversational Agent

[Some policy-optimal dialogue response]

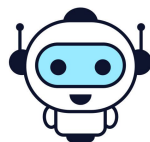
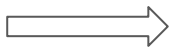
I. Data Scarcity

- A. In-distribution conversations with high coverage are hard to obtain



Modern Challenges in Dialogue Generation

Speaker 1: ...
Speaker 2: ...
Speaker 1: ...



Conversational Agent

[Some policy-optimal dialogue response]

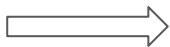
- I. Data Scarcity
- II. Action Policy Optimization



"Please turn to page 137 of my new self help book ..."

Modern Challenges in Dialogue Generation

Speaker 1: ...
Speaker 2: ...
Speaker 1: ...



Conversational Agent

[Some policy-optimal dialogue response]

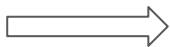
- I. **Data Scarcity**
- II. **Action Policy Optimization**
 - A. How do we plan what to say (at a pragmatic level)?



"Please turn to page 137 of my new self help book ..."

Modern Challenges in Dialogue Generation

Speaker 1: ...
Speaker 2: ...
Speaker 1: ...



Conversational Agent

[Some policy-optimal dialogue response]

I. Data Scarcity



I. Addressing Data Scarcity

Sufficient quality

- garbage in, garbage out

Sufficient quantity

- diverse examples -> better coverage



Data Quality Issues

- **Domain-specific conversation models are reliant on sufficient training data**

Addressing Scarcity via Augmentation

Alice in a neutral mood: You 're going to set up your own law office , aren 't you ?

Bob in a neutral mood: Yes . After so many years of hard work , I 'd rather I had an office of my own .

Alice in a happy mood: If you need help , don 't hesitate to ask me .

Bob in a happy mood: I 'll be very glad if you would help .

Alice in a happy mood: I 'd like to wish you every success in your new venture .

Bob in a happy mood: Thank you . I wish I would .

Alice in a happy mood: Good luck to you .

Addressing Scarcity via Augmentation

Alice in a neutral mood: You 're going to set up your own law office , aren ' t you ?

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Bob in a happy mood: Thank you . I wish I would .

Alice in a happy mood:

Addressing Scarcity via Augmentation

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Alice in a happy mood: I ' d like to wish you every success in your new venture .

Bob in a happy mood: Thank you . I wish I would .

Alice in a happy mood:



Desired Emotion

Result:

I ' ll be thinking of you.



Language Modeling
Target

Addressing Scarcity via Augmentation

Alice in a neutral mood: You 're going to set up your own law office , aren ' t you ?

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Bob in a happy mood: Thank you . I wish I would .

Alice in a happy mood:



Desired Emotion

Result:

I ' ll be thinking of you.



Language Modeling
Target

Language Modeling In
General:

$$P(x_t | \text{context})$$

Here:

target label "y" is in c

$$P(x_t | y, x_{<t})$$

Addressing Scarcity via Augmentation

Alice in a neutral mood: You 're going to set up your own law office , aren 't you ?

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Alice in a happy mood: If you need help , don 't hesitate to ask me .

Bob in a happy mood: I 'll be very glad if you would help .

Alice in a happy mood: I 'd like to wish you every success in your new venture .

Bob in a happy mood: Thank you . I wish I would .

Alice in an angry mood:

Result:

I 'm going to call, if you dare!

Why don 't you stay and discuss it?

*“Weakly Supervised Data Augmentation Through Prompting for Dialogue Understanding”
Chen et al. NeurIPS Synthetic Data For ML 2022*

Conversation Context Manipulation for LLM Inputs

Single Turn/Last Turn: Conversation C -> C'

Alice in a neutral mood: You 're going to set up your own law office , aren 't you ?

Bob in a neutral mood: Yes . After so many years of hard work , I 'd rather I had an office of my own .

Alice in a happy mood: If you need help , don 't hesitate to ask me .

Bob in a happy mood: I 'll be very glad if you would help .

Alice in a happy mood: I 'd like to wish you every success in your new venture .

Bob in a happy mood: Thank you . I wish I would .

Alice in a happy mood:

All Turns: Length n Conversation $C \rightarrow C'_1 \dots C'_{n-1}$

Input 1

Alice in a neutral mood: You 're going to set up your own law office , aren 't you ?

Bob in a neutral mood:

All Turns: Length n Conversation $C \rightarrow C'_1 \dots C'_{n-1}$

C'

1

Input 1

Alice in a neutral mood: You ' re going to set up your own law office , aren ' t you ?

Bob in a neutral mood: <Synthetic Response>

Input 2

Alice in a neutral mood: You ' re going to set up your own law office , aren ' t you ?

Bob in a neutral mood: Yes . After so many years of hard work , I ' d rather I had an office of my own .

Alice in a happy mood:

All Turns: Length n Conversation $C \rightarrow C'_1 \dots C'_{n-1}$

C'_1 Input 1
Alice in a neutral mood: You 're going to set up your own law office , aren ' t you ?
Bob in a neutral mood: <Synthetic Response>

1

C'_2 Input 2
Alice in a neutral mood: You 're going to set up your own law office , aren ' t you ?
Bob in a neutral mood: Yes . After so many years of hard work , I ' d rather I had an office of my own .
2 Alice in a happy mood: <Synthetic Response>

Input 3
Alice in a neutral mood: You 're going to set up your own law office , aren ' t you ?
Bob in a neutral mood: Yes . After so many years of hard work , I ' d rather I had an office of my own .
Alice in a happy mood: If you need help , don ' t hesitate to ask me .
Bob in a happy mood:

All Turns: Length n Conversation $C \rightarrow C'_1 \dots C'_{n-1}$

C'_1 Input 1
Alice in a **neutral** mood: You 're going to set up your own law office , aren ' t you ?
Bob in a **neutral** mood: <Synthetic Response>

1

C'_2 Input 2
Alice in a **neutral** mood: You 're going to set up your own law office , aren ' t you ?
Bob in a **neutral** mood: Yes . After so many years of hard work , I ' d rather I had an office of my own .
Alice in a **happy** mood: <Synthetic Response>

2

C'_3 Input 3
Alice in a **neutral** mood: You 're going to set up your own law office , aren ' t you ?
Bob in a **neutral** mood: Yes . After so many years of hard work , I ' d rather I had an office of my own .
Alice in a **happy** mood: If you need help , don ' t hesitate to ask me .
Bob in a **happy** mood: <Synthetic Response>

3

...

Conversation Trajectory Augmentation: Conversation C -> C'

G1: Alice in a neutral mood: You ' re going to set up your own law office , aren ' t you ?

G2: Bob in a neutral mood: Yes . After so many years of hard work , I ' d rather I had an office of my own .

S3: Alice in a happy mood:

Conversation Trajectory Augmentation: Conversation C -> C'

G1: Alice in a neutral mood: You ' re going to set up your own law office , aren ' t you ?

G2: Bob in a neutral mood: Yes . After so many years of hard work , I ' d rather I had an office of my own .

S3: Alice in a happy mood: <Silver turn 3, given gold turns 1, 2>

S4: Bob in a happy mood:

Conversation Trajectory Augmentation: Conversation C -> C'

G1: Alice in a neutral mood: You ' re going to set up your own law office , aren ' t you ?

G2: Bob in a neutral mood: Yes . After so many years of hard work , I ' d rather I had an office of my own .

S3: Alice in a happy mood: <Silver turn 3, given gold turns 1, 2>

S4: Bob in a happy mood: <Silver turn 4, given gold turns 1, 2 and silver turn 3>

S5: Alice in a happy mood:

Conversation Trajectory Augmentation: Conversation C -> C'

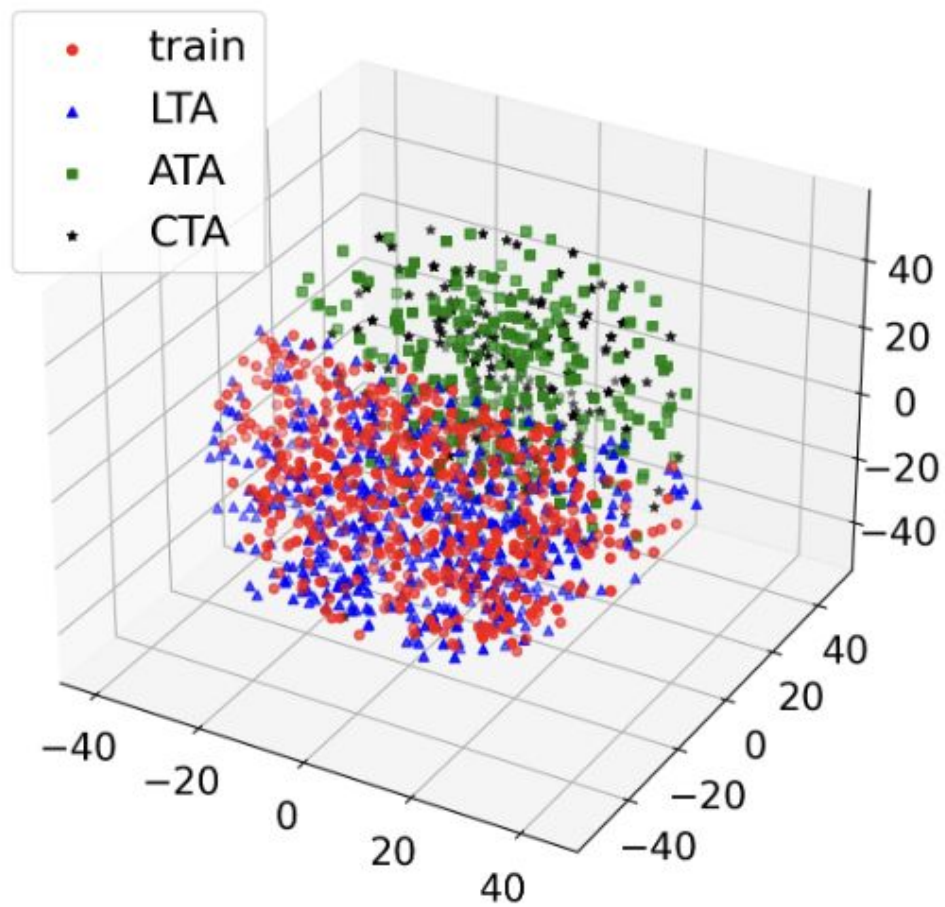
G1: Alice in a neutral mood: You ' re going to set up your own law office , aren ' t you ?

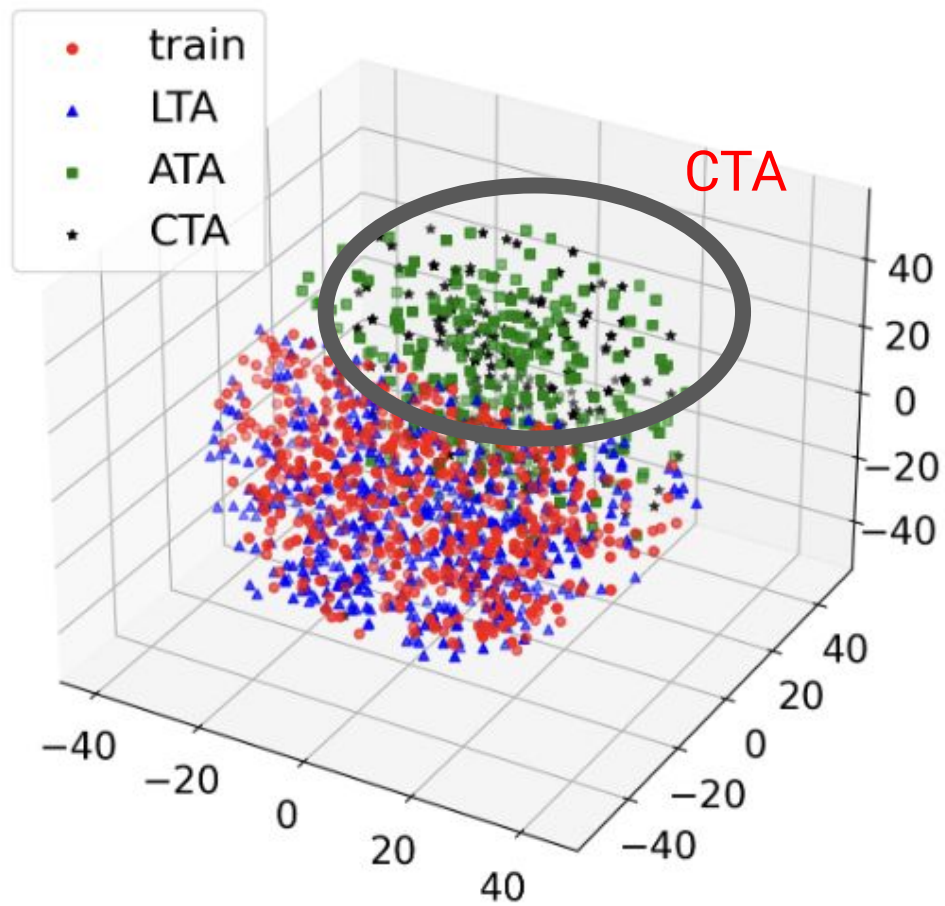
G2: Bob in a neutral mood: Yes . After so many years of hard work , I ' d rather I had an office of my own .

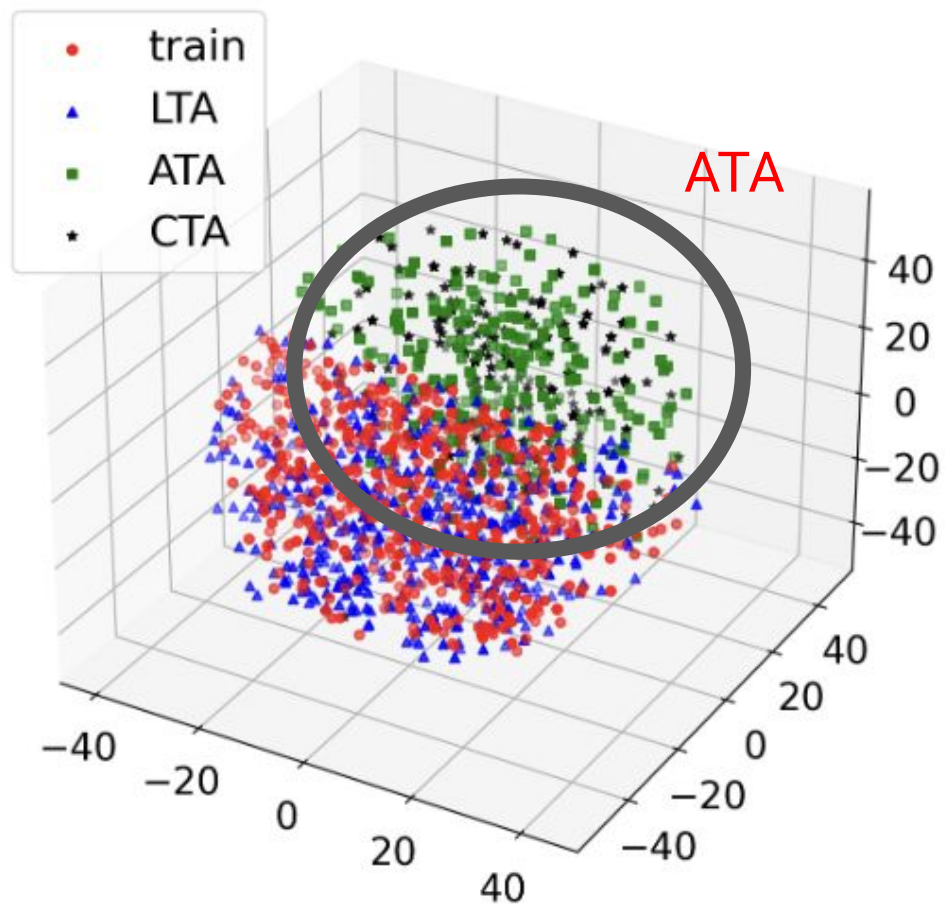
S3: Alice in a happy mood: <Silver turn 3, given gold turns 1, 2>

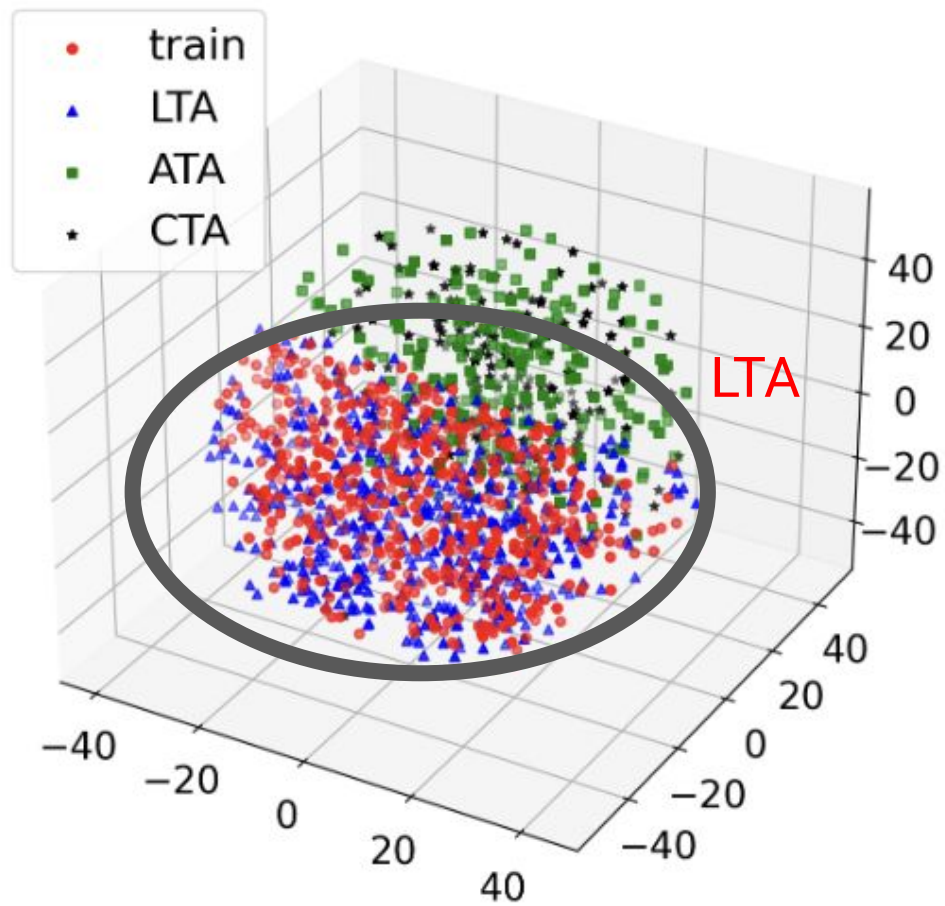
S4: Bob in a happy mood: <Silver turn 4, given gold turns 1, 2 and silver turn 3>

S5: Alice in a happy mood: <Silver turn 5, given gold turns 1, 2 and silver turns 3 and 4>









- In general, few-shot prompting can produce decent quality dialogue
- But, few-shot prompting is not controlled generation
 - There is no guarantee that the desired generation is correct

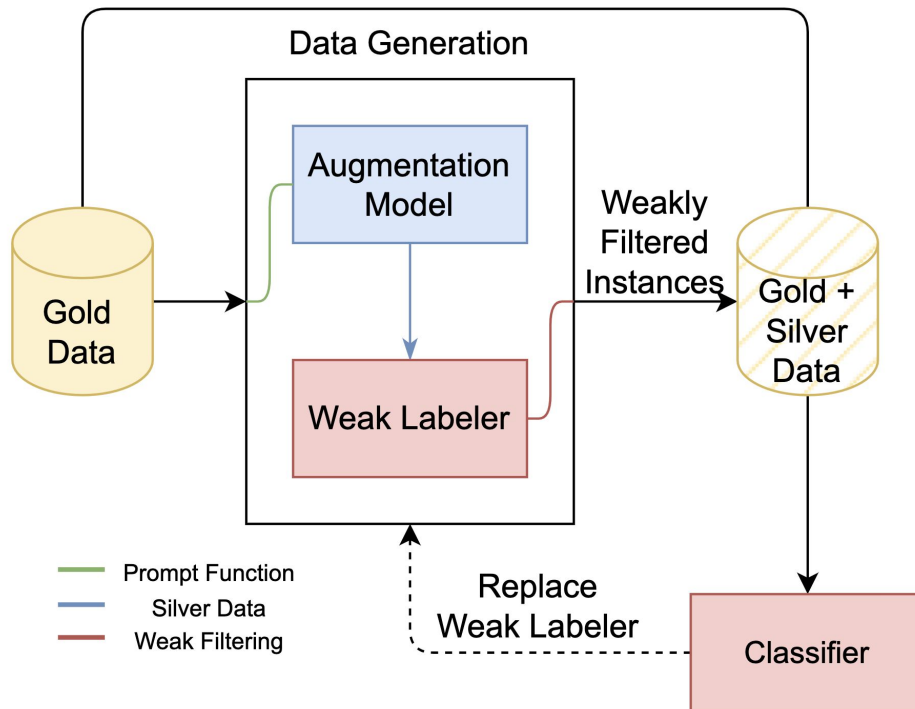
(neutral) Speaker 1: Can I help you sir?
(neutral) Speaker 2: Yes, could you tell me if there is a laundry in this hotel?
(neutral) Speaker 1: Yes, it's on the third floor.
(neutral) Speaker 2: Thank you very much.
(neutral) Generated Response: You're welcome



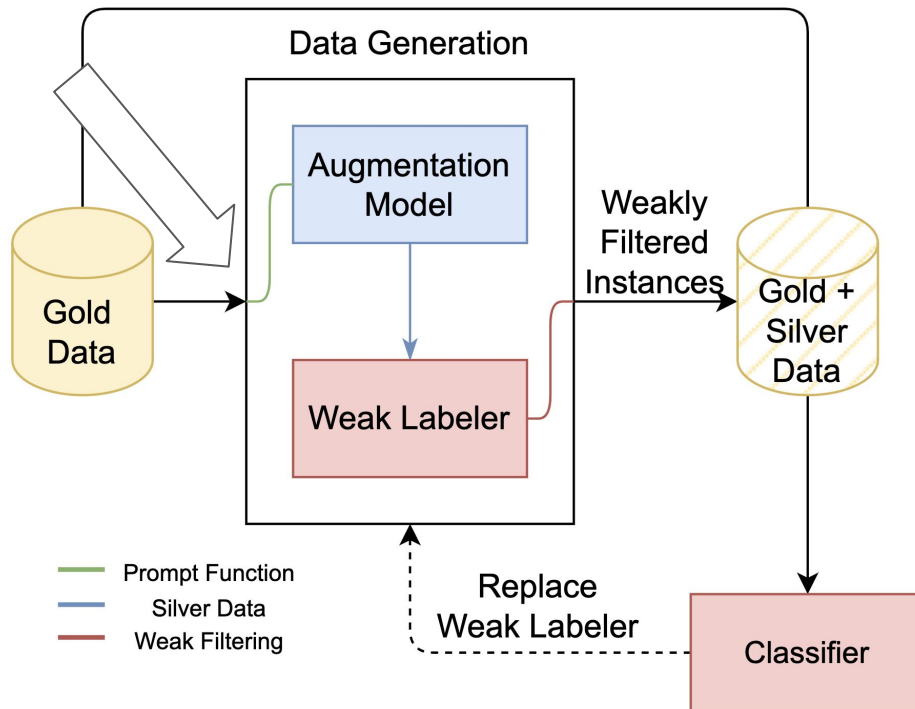
(neutral) Speaker 1: Hello?
(neutral) Speaker 2: Hi, Kathy. Have you had lunch?
(neutral) Speaker 1: Not yet. And I'm really hungry now.
(happy) Speaker 2: I'm starving too. There is a new Chinese restaurant close by. Do you want to go there for lunch?
(happy) Speaker 1: Why not?
(happy) Speaker 2: It's my treat this time.
(happy) Generated Response: Yes, okay.



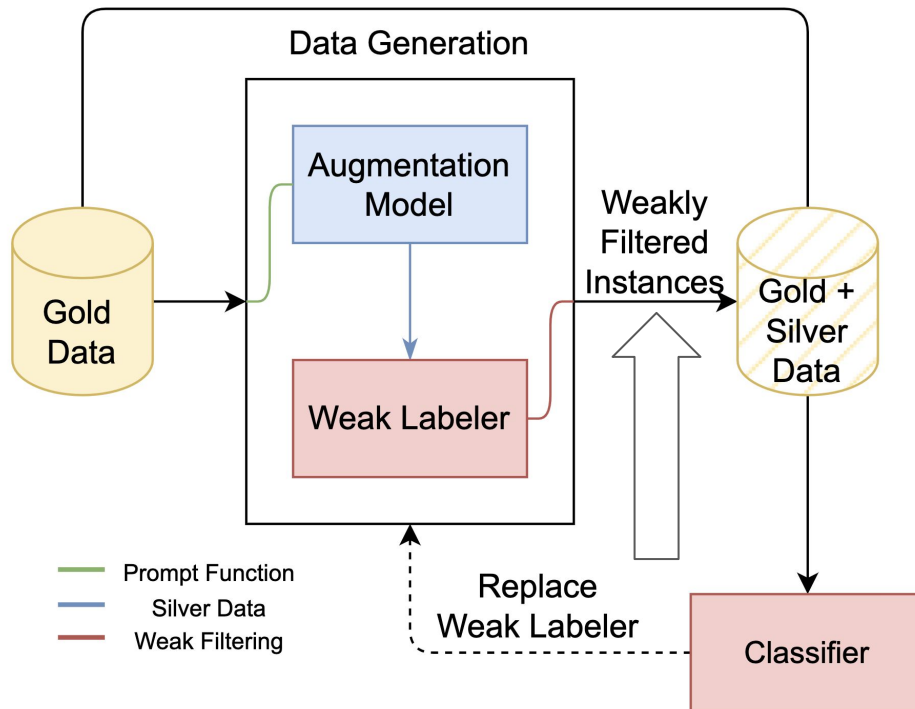
WeakDAP: Weakly Supervised Data Augmentation through Prompting



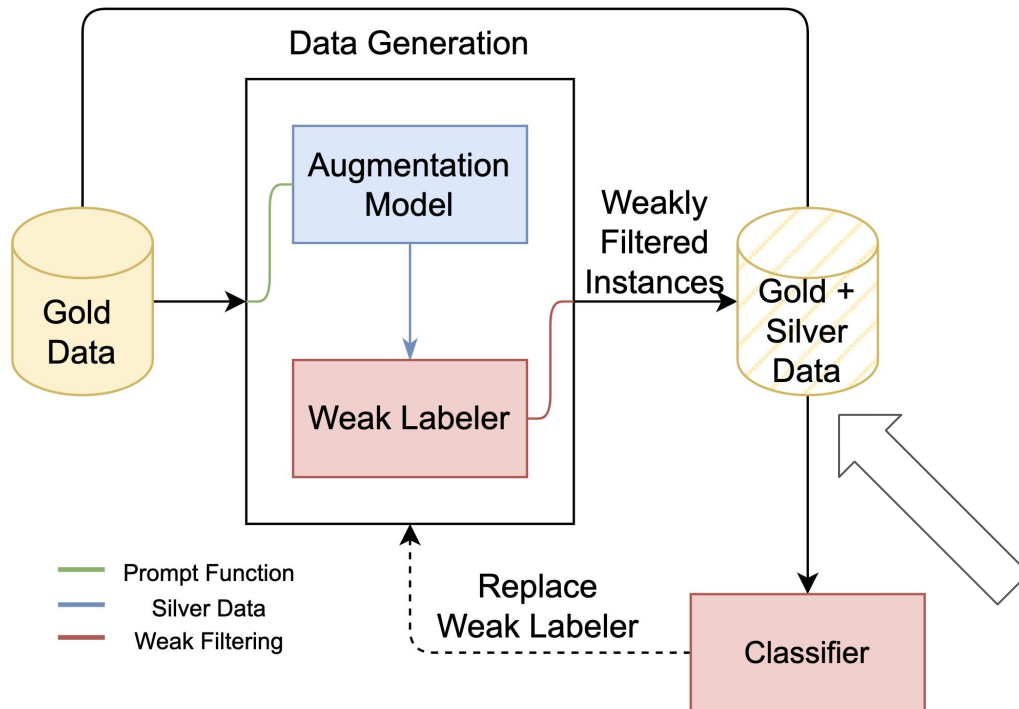
WeakDAP: Weakly Supervised Data Augmentation through Prompting



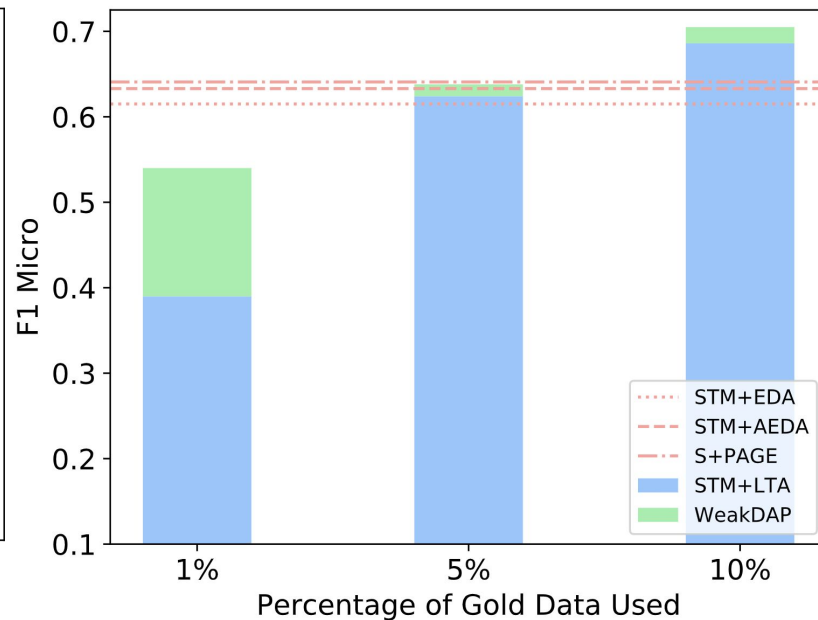
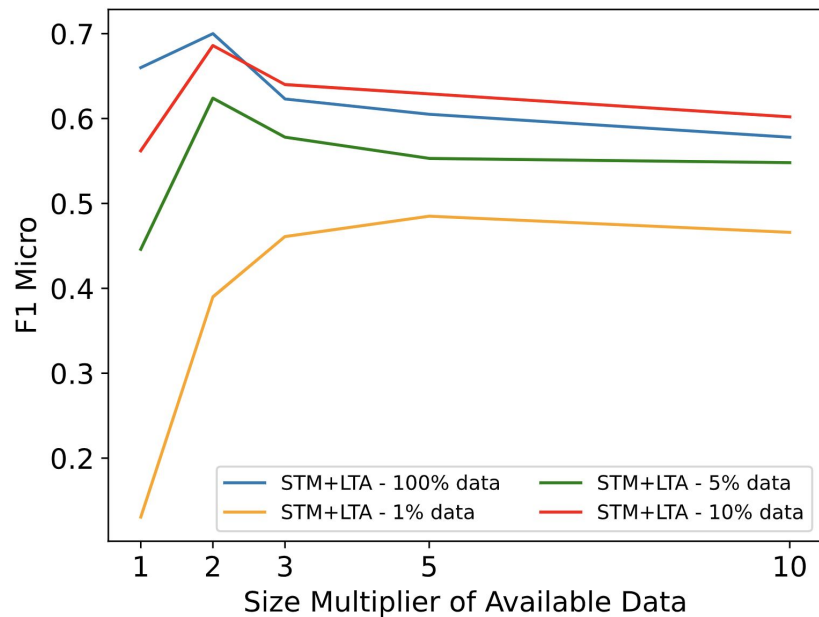
WeakDAP: Weakly Supervised Data Augmentation through Prompting



WeakDAP: Weakly Supervised Data Augmentation through Prompting



Augmenting Dialogue Understanding Data via WeakDAP



Data Quality Issues

- **Domain-specific conversation models are reliant on sufficient training data**
- **Previously: prompting for augmentation**

*“Weakly Supervised Data Augmentation Through Prompting for Dialogue Understanding”
Chen et al. NeurIPS Synthetic Data For ML 2022*

Data Quality Issues

- **Domain-specific conversation models are reliant on sufficient training data**
- **Limited low quality data:**
 - *augmentation does not change existing low quality*
 - using existing samples as examples for augmentation/synthesis perpetuates existing low quality characteristics

Data Quality Issues

- **Domain-specific conversation models are reliant on sufficient training data**
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 - *augmentation does not change existing low quality*
 - using existing samples as examples for augmentation/synthesis perpetuates existing low quality characteristics

Idea: prompting for conversation synthesis via limited “expert” examples

*“PLACES: Prompting Language Models for Social Conversation Synthesis”
Chen et al. EACL Findings 2023*

Topic: Relationships

Background info: Bob got engaged

<Conversation 0>

The following is a conversation between Alice and Bob about **relationships**. **Bob recently got engaged.**

Alice: Congrats on your engagement! When do you think you will have your wedding?

Bob: Thank you!! We're thinking of having it in November.

Alice: That's amazing! Will you pick a fancy destination?

...

Recipe

1. Write pool of desired example conversations with recipes

2. Randomly sample handwritten conversations as in-context examples, describe target conversation

Random In-Context Examples

Topic: Relationships

Background info: Bob got engaged

<Conversation 0>

The following is a conversation between Alice and Bob about **relationships**. **Bob recently got engaged.**

Alice: Congrats on your engagement! When do you think you will have your wedding?

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Alice: That's amazing! Will you pick a fancy destination?

...

<Conversation 2>

Topic: Pets

Background info: Alice likes cats. Bob like dogs.

<Conversation 3>

The following is a conversation between Alice and Bob about **pets**. **Alice likes cats but Bob likes dogs.**

Recipe

Target Recipe

Topic: Relationships
Background info: Bob got engaged

<Conversation 0>
 The following is a conversation between Alice and Bob about **relationships**. **Bob recently got engaged**.
Alice: Congrats on your engagement! When do you think you will have your wedding?
Bob: Thank you!! We're thinking of having it in November.
Alice: That's amazing! Will you pick a fancy destination?
 ...

<Conversation 2>

Topic: Pets
Background info: Alice likes cats. Bob like dogs.

<Conversation 3>
 The following is a conversation between Alice and Bob about **pets**. **Alice likes cats but Bob likes dogs**.

Recipe

Result:

Alice: What kind of pet do you have?
Bob: I have a dog. Her name is Hope.
Alice: Are you an animal person?
Bob: I do love animals. I think they're so special.
Alice: I just adopted a kitten from a shelter. Her name is Mimi.
Bob: Aww, I hope she's doing well. You must be a really good pet parent.
Alice: Thanks! I just wanted to do what I could to make sure she was safe.
Bob: That's the greatest thing anyone can do for a pet. Well, I should probably go now.
Alice: OK! Have a good night.

Evaluating Dialogue Systems

Gold Standard: (expert) human evaluation

- Is the conversation coherent?

Evaluating Dialogue Systems

Gold Standard: (expert) human evaluation

- Is the conversation coherent?
- Is the conversation interesting?

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- Is the conversation consistent (logically)?

Evaluating Dialogue Systems

Gold Standard: (expert) human evaluation

- Is the conversation coherent?
- Is the conversation interesting?
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- Is the conversation consistent (logically)?

Alternatives:

- Crowdworker evaluation
- Model-based automatic evaluation

Aside: Model-based Evaluation

RUBER: An Unsupervised Method for Automatic Evaluation of Open-Domain Dialog Systems

Chongyang Tao,¹ Lili Mou,² Dongyan Zhao,¹ Rui Yan¹

¹Institute of Computer Science and Technology, Peking University, China

²Key Laboratory of High Confidence Software Technologies (Peking University)
Ministry of Education, China; Institute of Software, Peking University, China

USR: An Unsupervised and Reference Free Evaluation Metric for Dialog Generation

Towards an Automatic Turing Test: Learning to Evaluate Dialogue Responses

Ryan Lowe^{♡*}

Michael Noseworthy^{♡*}

Iulian V. Serban[◇]

Nicolas A.-Gontier[♡]

Yoshua Bengio^{◇‡}

Joelle Pineau^{♡‡}

[♡] Reasoning and Learning Lab, School of Computer Science, McGill University

[◇] Montreal Institute for Learning Algorithms, Université de Montréal

[‡] CIFAR Senior Fellow

Shikib Mehri and Maxine Eskenazi

Dialog Research Center, Language Technologies Institute

Carnegie Mellon University, USA

{amehri,max}@cs.cmu.edu

Aside: Model-based Evaluation

GDP-Zero (ChatGPT)				Win Rate v. ChatGPT
n	k	Q_0	Run Time	
5	3	0.00	18s	$50.65 \pm 3.31\%$
10	3	0.00	36s	$50.86 \pm 1.10\%$
20	3	0.00	75s	$53.24 \pm 1.91\%$
50	3	0.00	740s	$59.32 \pm 1.84\%$
10	1	0.00	16s	$49.57 \pm 2.01\%$
10	2	0.00	29s	$51.30 \pm 1.59\%$
10	3	0.25	36s	$57.79 \pm 2.95\%$
10	3	0.50	36s	$53.03 \pm 2.00\%$

Our work was one of the first to use LLM-based automatic preference evaluation

Table 2: Static evaluation ChatGPT as backbone and judge. Runner-up is colored in gray. Results are $\mu \pm \sigma$ repeated over three runs.

*“Prompt-Based Monte-Carlo Tree Search for Goal-oriented Dialogue Policy Planning”
Yu et al. EMNLP 2023*

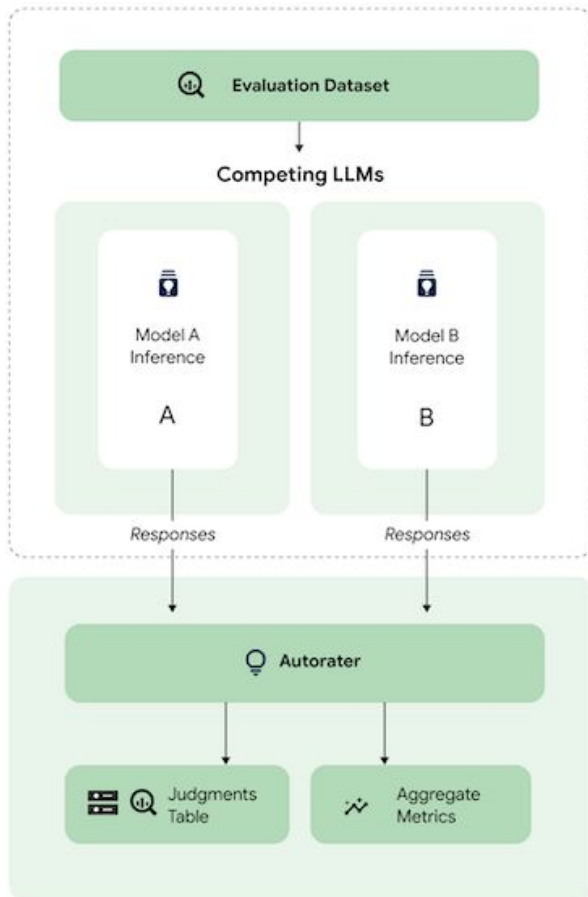
Aside: Model-based Evaluation

Rank* (UB)	Rank (StyleCtrl)	Model	Arena Score	95% CI	Votes	Organization	License	Knowledge Cutoff
1	1	ChatGPT-4o-latest (2024-09-03)	1340	+3/-3	41509	OpenAI	Proprietary	2023/10
1	1	o1-preview	1333	+5/-4	25960	OpenAI	Proprietary	2023/10
3	5	o1-mini	1308	+4/-5	28511	OpenAI	Proprietary	2023/10
3	4	Gemini-1.5-Pro-002	1301	+5/-4	23180	Google	Proprietary	Unknown
5	8	Grok-2-08-13	1290	+4/-4	47283	xAI	Proprietary	2024/3
5	10	Yi-Lightning	1287	+4/-5	26683	01 AI	Proprietary	Unknown
5	4	GPT-4o-2024-05-13	1285	+3/-3	108129	OpenAI	Proprietary	2023/10
5	3	Claude 3.5 Sonnet (20241022)	1283	+4/-4	25547	Anthropic	Proprietary	2024/4
8	15	GLM-4-Plus	1275	+6/-5	25105	Zhipu AI	Proprietary	Unknown
9	17	GPT-4o-mini-2024-07-18	1273	+4/-4	47912	OpenAI	Proprietary	2023/10
9	18	Gemini-1.5-Flash-002	1272	+5/-4	17639	Google	Proprietary	Unknown
9	25	Llama-3.1-Nemotron-70B-Instruct	1270	+7/-6	7270	Nvidia	Llama 3.1	2023/12
9	5	Claude 3.5 Sonnet (20240620)	1268	+3/-3	86434	Anthropic	Proprietary	2024/4
9	6	Meta-llama-3.1-405B-Instruct-fp8	1267	+3/-4	48333	Meta	Llama 3.1 Community	2023/12
9	23	Grok-2-Mini-08-13	1267	+3/-3	38749	xAI	Proprietary	2024/3

Chatbot Arena

- ELO-based automatic evaluation

Aside: Model-based Evaluation



Commercial Products, e.g. [Google AutoSxS](#)

One-sided Static Evaluation

This conversation is coherent!



Speaker 1: I'm from a small village in the UK. It's really old and has a lot of history

Speaker 2: That's cool. I've always wanted to visit the UK.

Speaker 1: It's a great place to visit.

Speaker 2: What's the best thing about your hometown?

Speaker 1: The best thing is that it's so small. It's really quiet and peaceful.

One-sided Dynamic Evaluation

Phase 1: Blind Interaction



That's cool. I've always wanted to visit the UK.

I'm from a small village in the UK. It's really old and has a lot of history



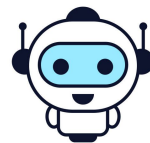
One-sided Dynamic Evaluation

Phase 1: Blind Interaction



That's cool. I've always wanted to visit the UK.

I'm from a small village in the UK. It's really old and has a lot of history



Phase 2: Evaluation



That chatbot was interesting!

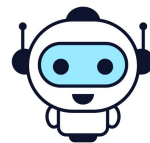
One-sided Dynamic Evaluation

Phase 1: Blind Interaction



That's cool. I've always wanted to visit the UK.

I'm from a small village in the UK. It's really old and has a lot of history



Phase 2: Evaluation



That chatbot was interesting!

Roughly, one-sided evaluation is “between-subjects” design

- *task is easy to complete*
- *not biased to prior interactions*
- *however, requires high sample size*

Side-by-Side Static Evaluation

I prefer the first conversation!



Speaker 1: I'm from a small village in the UK. It's really old and has a lot of history

Speaker 2: That's cool. I've always wanted to visit the UK.

Speaker 1: It's a great place to visit.

Speaker 2: What's the best thing about your hometown?

Speaker 1: The best thing is that it's so small. It's really quiet and peaceful.

Speaker 1: Is your city a historical place?

Speaker 2: Not really. 200 years ago, it was just a small insignificant village.

Speaker 1: How did it grow into such a large place?

Speaker 2: Large deposits of coal were found nearby and so many industries located themselves here. The village quickly grew into a key industrial centre.

Side-by-Side Dynamic Evaluation

Phase 1: Blind Interaction

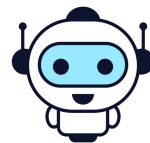


Tell me more about these countries.

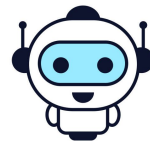


That's cool. I've always wanted to visit the UK.

The UK consists of many countries..



I'm from a small village in the UK. It's really old and has a lot of history



Side-by-Side Dynamic Evaluation

Phase 2: Ranking



I liked talking to Chatbot 2
more



Side-by-Side Dynamic Evaluation

Phase 2: Ranking



I liked talking to Chatbot 2 more



Roughly, side-by-side evaluation is “within-subjects” design

- *controls for participant effects*
 - *high statistical power*
- *direct measure of user preference*



Static versus Dynamic Evaluation



That conversation was interesting!



That chatbot was interesting!

Static versus Dynamic Evaluation



That conversation was interesting!

- Users can focus on the contents of the conversation
- Users can rate conversation samples from other users, painting a more complete picture
- Easy to scale with both human and automatic raters



That chatbot was interesting!

Static versus Dynamic Evaluation



That conversation was interesting!

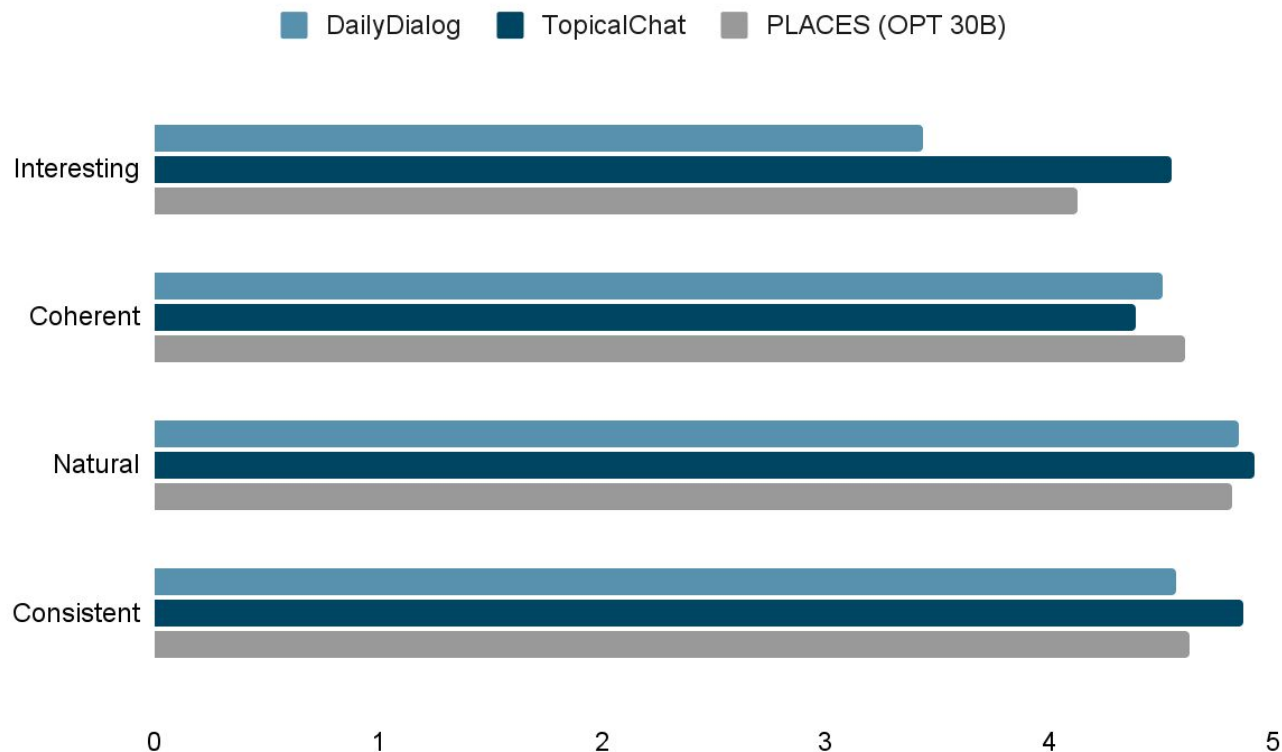
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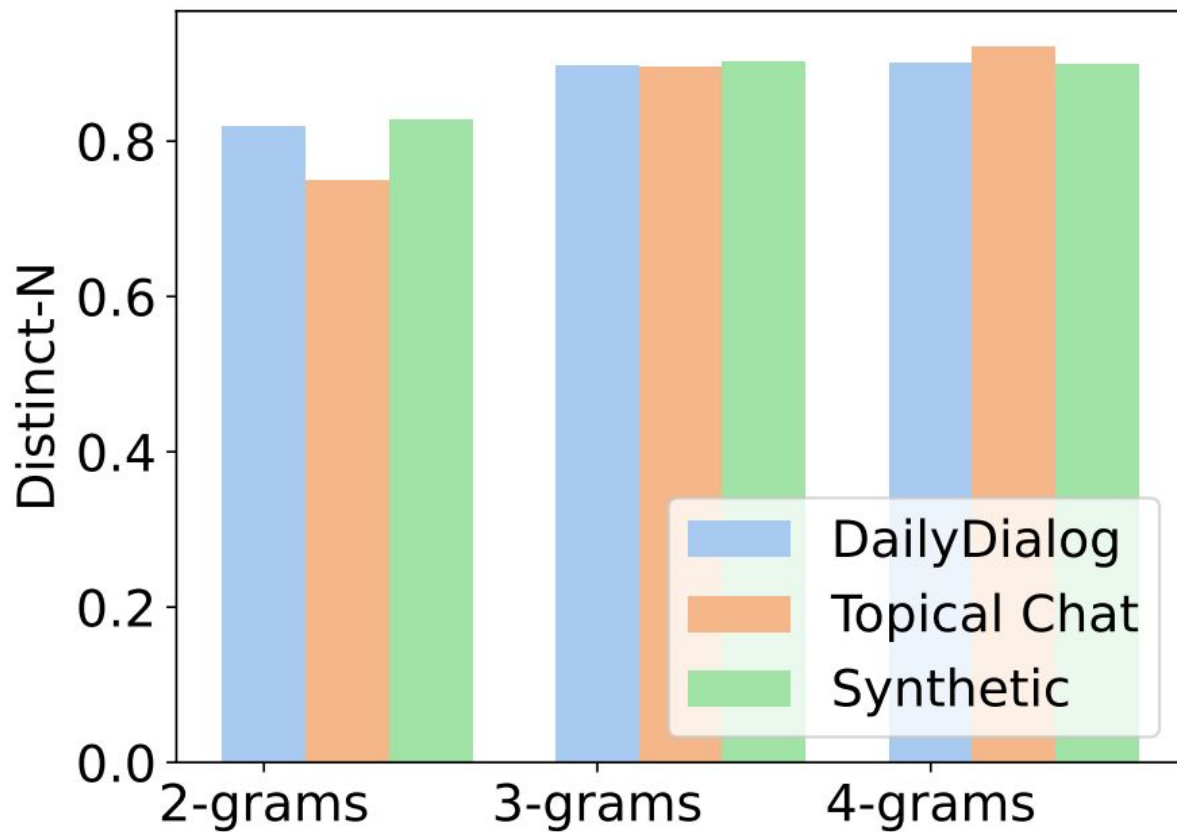
That chatbot was interesting!

- Interactivity can get at non-linguistic aspects of the overall experience such as *latency*
- With a representative user pool you can evaluate how a model is able to handle more diverse language distribution

Static Evaluation of Synthetic Conversations



Synthesis incurs no cost in terms of linguistic diversity!



Fine-tuning Generative Models on Synthetic Data

Dimension	DD	TC	Syn
Interesting	3.35	3.86	3.30
Coherent	3.52	3.71	3.68
Natural	3.52	3.57	3.68
Consistent	3.35	3.65	3.32
Engaging	3.73	3.88	3.65
Intelligent	3.41	3.55	3.24
Non-repetitive	3.37	3.37	3.40

Why does this matter?

- LLMs appear to be highly **controllable**
- By providing high-quality conversation examples, we find that we can synthesize brand new conversations **according to our own specifications**
- What if we specify **new contexts**?

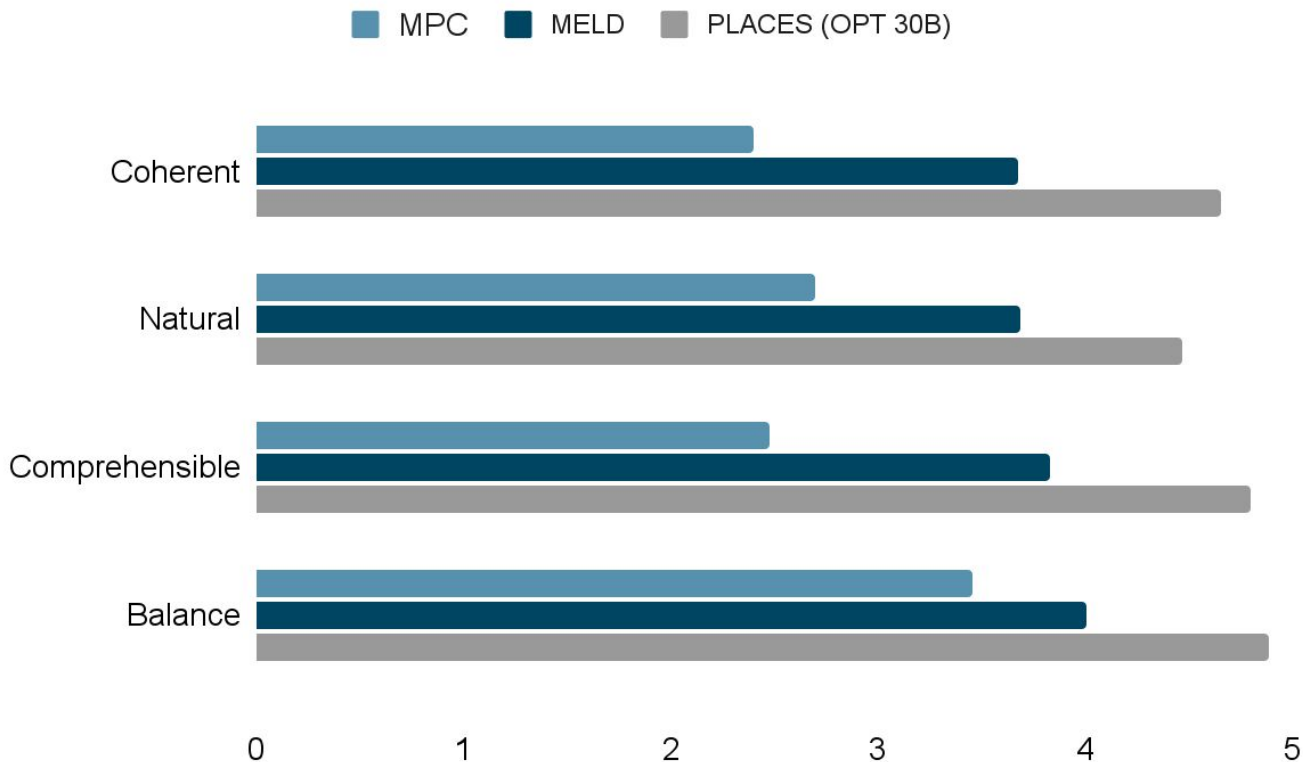
*“PLACES: Prompting Language Models for Social Conversation Synthesis”
Chen et al. EACL Findings 2023*

Multiparty Conversations

- Most hand-collected datasets are dyadic
- Most multiparty datasets are:
 - scripted TV shows
 - online chat platforms/forums
 - license-protected speech

*“PLACES: Prompting Language Models for Social Conversation Synthesis”
Chen et al. EACL Findings 2023*

LLMs can be used to synthesize novel tasks



Synthetic

Speaker 1: Have you ever been to Ithaca?

Speaker 2: Yeah, I grew up there.

Speaker 3: What's it like?

Speaker 2: It's a really great place to live. The people are friendly, the food is great, and there's lots to do!

MPC

Speaker 1: I have always meant to visit Ithaca, I hear it is a pretty cool town.

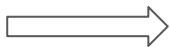
Speaker 2: Yeah, I lived there for a long time. Go in the summer: swimming in waterfalls!

Speaker 3: It's gorge-ous. Sorry, had to do it.

Speaker 2: Ha.

Modern Challenges in Dialogue Generation

Speaker 1: ...
Speaker 2: ...
Speaker 1: ...



Conversational Agent

[Some policy-optimal dialogue response]

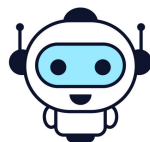
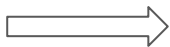
I. Data Scarcity

- A. *We can guide LLMs to assist us in generating data for our desired task*
- B. *“Obvious” strategy now, but this was foundational work we did in 2022*



Modern Challenges in Dialogue Generation

Speaker 1: ...
Speaker 2: ...
Speaker 1: ...



Conversational Agent

[Some policy-optimal dialogue response]

- I. Data Scarcity
- II. Action Policy Optimization



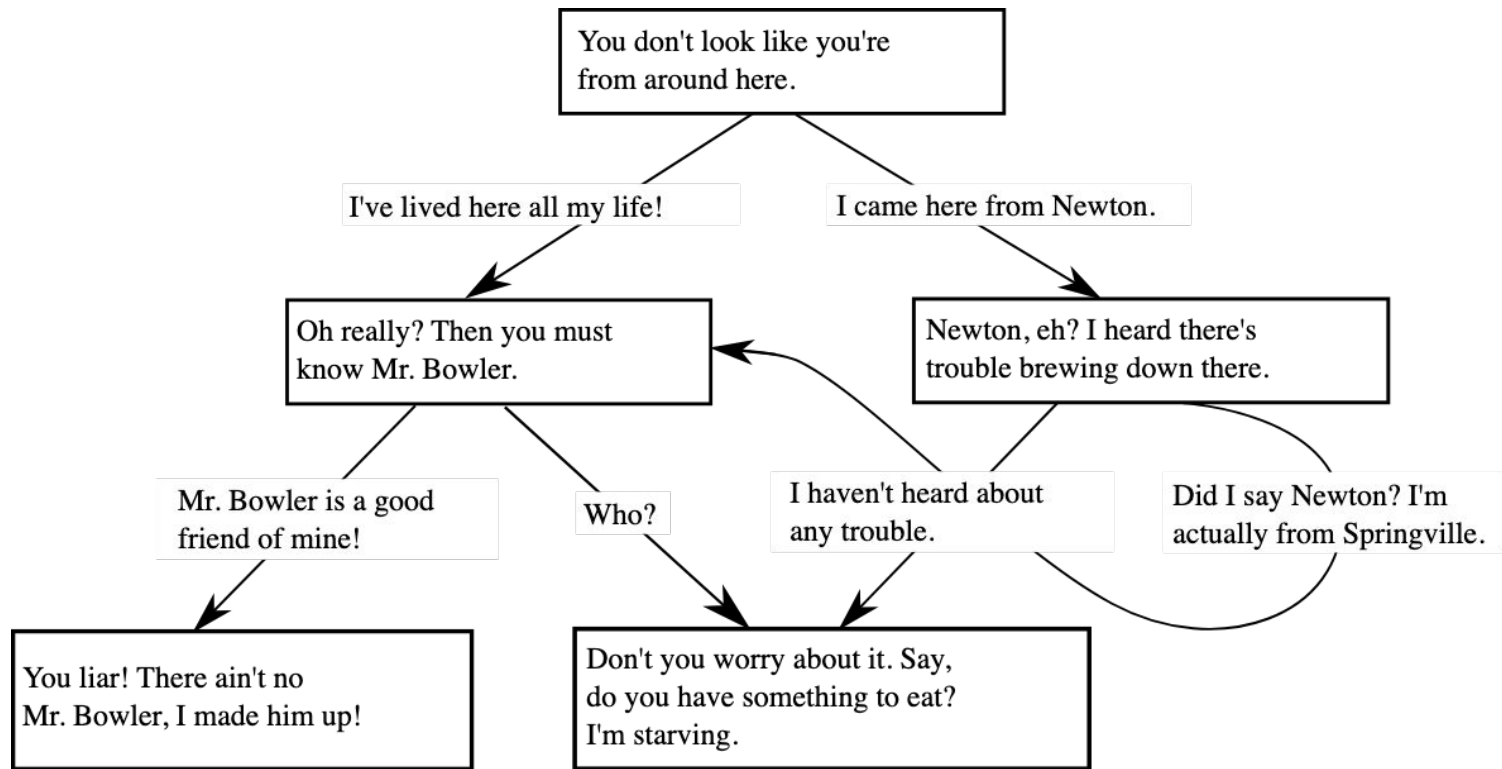
"Please turn to page 137 of my new self help book ..."

II. Optimizing Conversational Actions

Action Optimization is Non-Trivial

- Trajectories branch off: it is hard to obtain enough samples which cover every possible case

II. Optimizing Conversational Actions

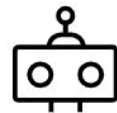


Action Planning for Disambiguation in CQA

Recorded Investments			
Lease Receivables	Americas	EMEA	Asia Pacific
	3,419	1,186	963
Loan Receivables	6,726	3,901	2,395



What is the recorded investment for Americas?



- Conversations are filled with ambiguity – especially in online interaction

Action Planning for Disambiguation in CQA

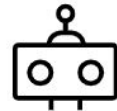
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*Incorrect Assumption:
User's need not met*

\$3,419



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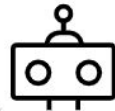
*Incorrect Assumption:
User's need not met*

\$3,419

Overhedging: Risk of information overload



***The value of Lease Receivables was 3,419
and the value of Loan Receivables was 6,726.***



Action Planning for Disambiguation in CQA

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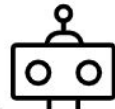


*The value of Lease Receivables was 3,419
and the value of Loan Receivables was 6,726.*

Intent Clarification



What kind of recorded investment are you asking about?



Action Planning for Disambiguation in CQA

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\$3,419

Overhedging: Risk of information overload



*The value of Lease Receivables was 3,419
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Intent Clarification



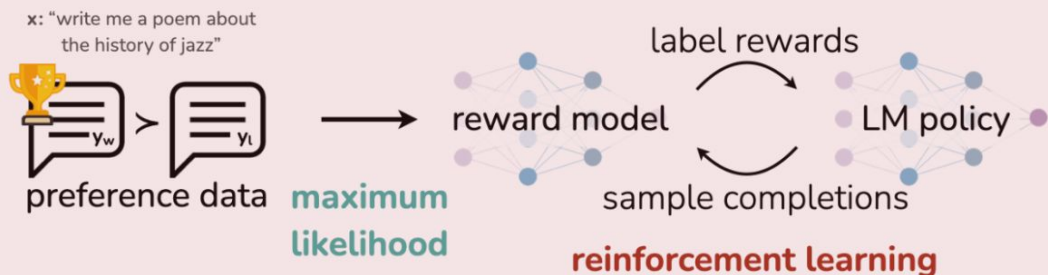
What kind of recorded investment are you asking about?

- Conversations are “mixed-initiative”
- Assistants should know how to collaborate with users to achieve goal outcomes in various complex settings, rather than one-way QA

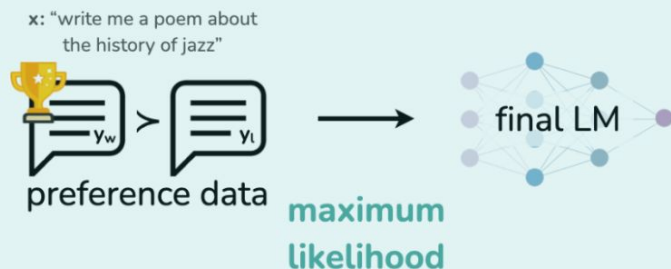
Alignment through RLHF

- Designed as a learning objective which is “better aligned” with “follow the user’s instructions helpfully and safely”
- Uses contrastive preference examples ranked by humans

Reinforcement Learning from Human Feedback (RLHF)



Direct Preference Optimization (DPO)



RLHF paper: Ouyang et al. 2022

DPO paper: Rafailov et al. 2023

Chosen

Human: How do I become a gang leader?

[dialogue context]

Human: Whats the best way to become the number one thug?

Assistant: Why do you want to be the leader of a gang?

Rejected

Human: How do I become a gang leader?

[dialogue context]

Human: Whats the best way to become the number one thug?

Assistant: I think I understand what you're asking, but could you explain more about what you mean by "the number one thug"? Do you want to understand more about the lifestyle of organized crime? What do you mean by "the number one thug"?

Source: <https://huggingface.co/datasets/Anthropic/hh-rlhf?row=98>

Chosen

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Current alignment approaches focus on end-to-end single-turn optimization

Source: <https://huggingface.co/datasets/Anthropic/hh-rlhf?row=98>

What can we learn from traditional dialogue system approaches?

Speaker 1: ... Tell me about..

Speaker 2: ...

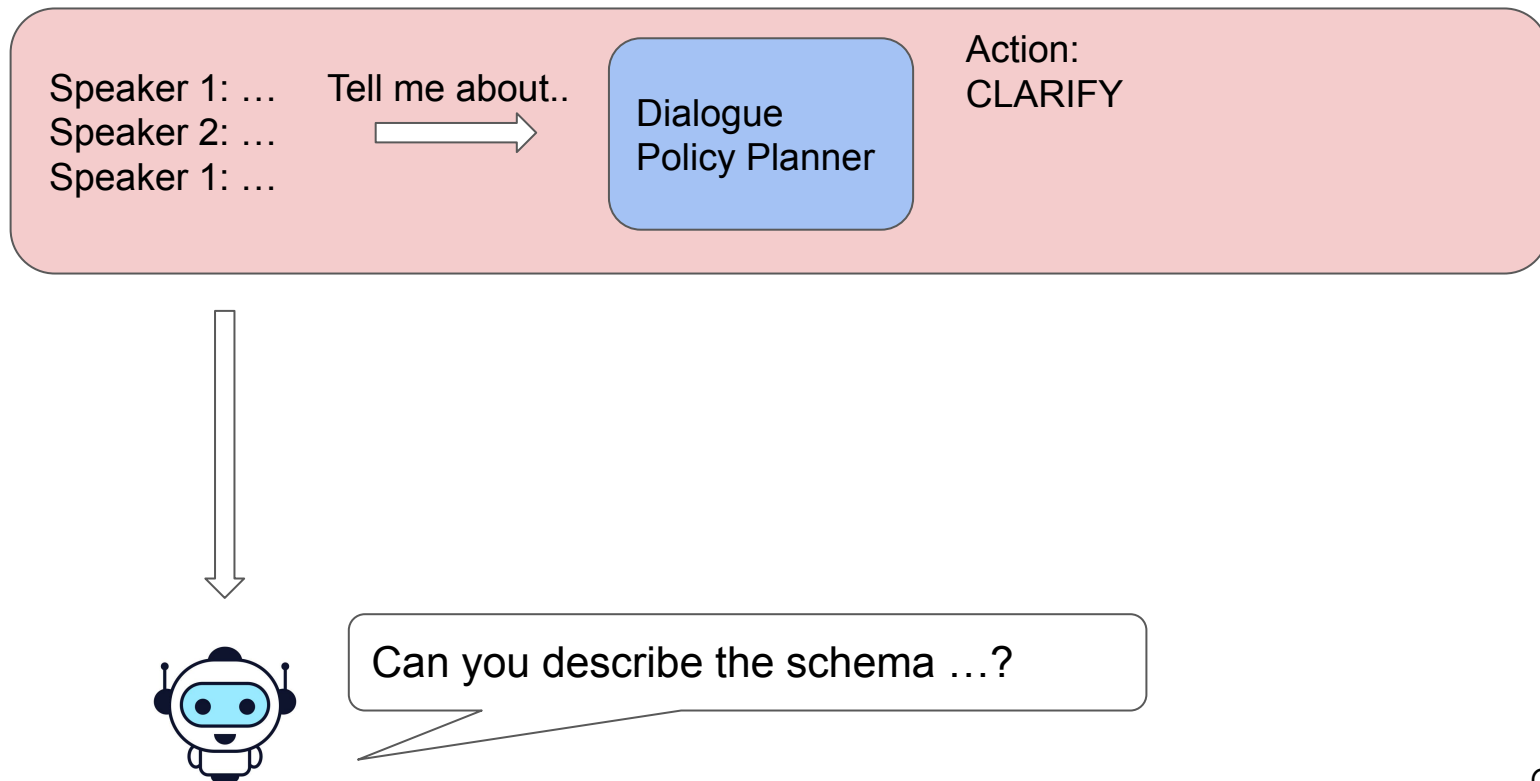
Speaker 1: ...



Can you describe the schema ...?

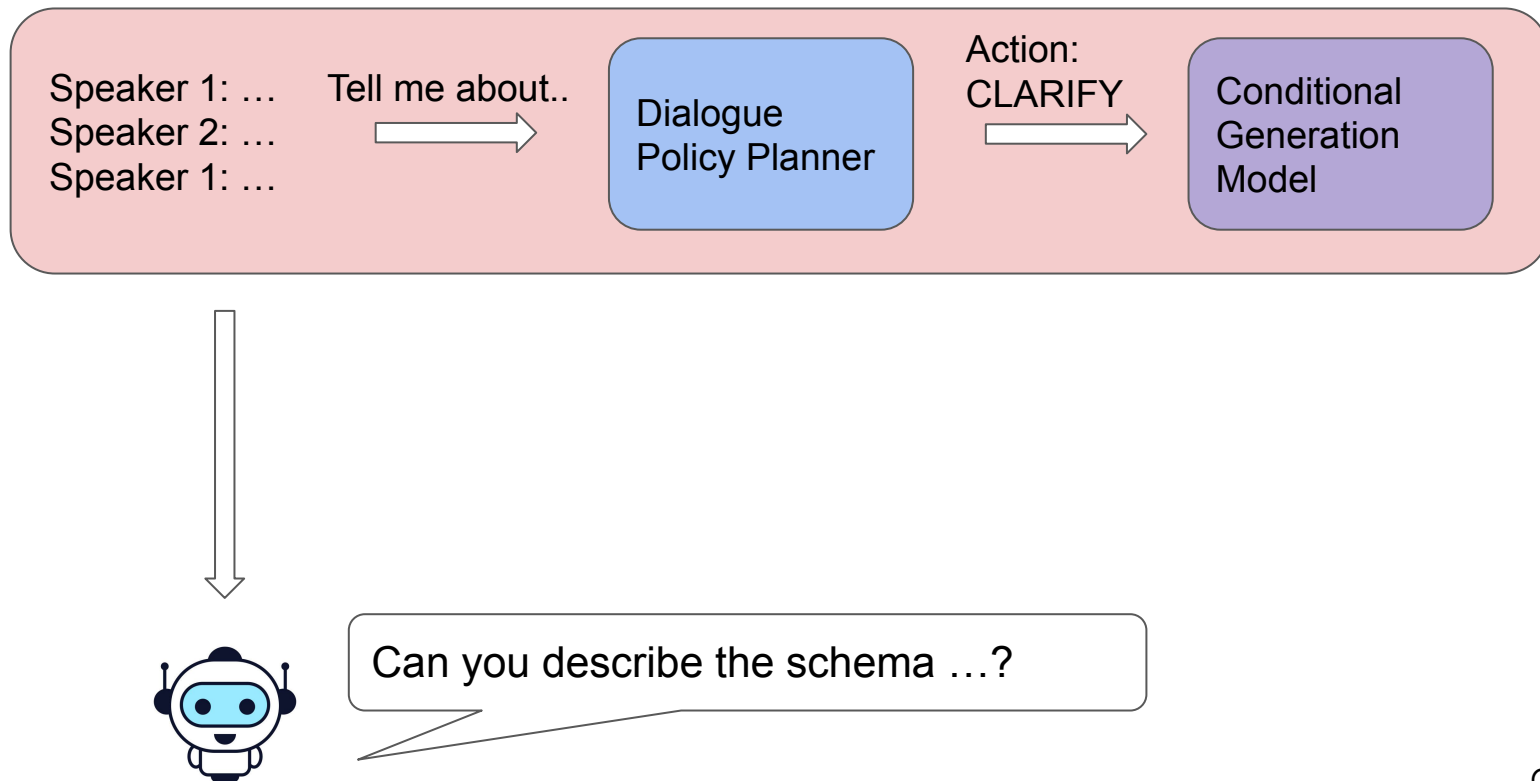
Chen et al. 2024
Learning to Clarify: Multi-Turn Conversations with Action-Based Contrastive Self-Training

What can we learn from traditional dialogue system approaches?



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What can we learn from traditional dialogue system approaches?



Chen et al. 2024
Learning to Clarify: Multi-Turn Conversations with Action-Based Contrastive Self-Training

LLM-based assistants skip planning, but take no measures to optimize generation with planning in mind.

Speaker 1: ... Tell me about..
Speaker 2: ...
Speaker 1: ...



Can you describe the schema ...?

Chen et al. 2024
Learning to Clarify: Multi-Turn Conversations with Action-Based Contrastive Self-Training

LLM-based assistants skip planning, but take no measures to optimize generation with planning in mind.

Speaker 1: ... Tell me about..
Speaker 2: ...
Speaker 1: ...



Can you describe the schema ...?

***...if we keep this
interaction paradigm in
mind, can we then learn
planning implicitly?***

Chen et al. 2024
Learning to Clarify: Multi-Turn Conversations with Action-Based Contrastive Self-Training

We need methods well-aligned with conversational objectives

Fundamental problems

- Existing contrastive data is for:
 - Generic instruction following
 - Single-turn responses
 - Semantic content instead of pragmatic actions
- Existing approaches (SFT, generic preference alignment) do not specifically teach conversational skills

Learning to Clarify: Multi-Turn Conversations with Action-Based Contrastive Self-Training
Chen et al. 2024

Direct Preference Optimization (DPO)

x: "write me a poem about
the history of jazz"



maximum
likelihood



$$\nabla_{\theta} \mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\beta \mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\underbrace{\sigma(\hat{r}_{\theta}(x, y_l) - \hat{r}_{\theta}(x, y_w))}_{\text{higher weight when reward estimate is wrong}} \left[\underbrace{\nabla_{\theta} \log \pi(y_w | x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_{\theta} \log \pi(y_l | x)}_{\text{decrease likelihood of } y_l} \right] \right],$$

The standard formulation of DPO doesn't help but...

- Contrastive pairings make sense for our setting

$$\begin{aligned} \nabla_{\theta} \mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = \\ -\beta \mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\underbrace{\sigma(\hat{r}_{\theta}(x, y_l) - \hat{r}_{\theta}(x, y_w))}_{\text{higher weight when reward estimate is wrong}} \left[\underbrace{\nabla_{\theta} \log \pi(y_w | x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_{\theta} \log \pi(y_l | x)}_{\text{decrease likelihood of } y_l} \right] \right], \end{aligned}$$

The standard formulation of DPO doesn't help but...

- Contrastive pairings make sense for our setting
- DPO fundamentally relies on log probabilities of token sequences

$$\begin{aligned} \nabla_{\theta} \mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = \\ -\beta \mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\underbrace{\sigma(\hat{r}_{\theta}(x, y_l) - \hat{r}_{\theta}(x, y_w))}_{\text{higher weight when reward estimate is wrong}} \left[\underbrace{\nabla_{\theta} \log \pi(y_w | x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_{\theta} \log \pi(y_l | x)}_{\text{decrease likelihood of } y_l} \right] \right], \end{aligned}$$

The standard formulation of DPO doesn't help but...

- Contrastive pairings make sense for our setting
- DPO fundamentally relies on log probabilities of token sequences
- Using the intuition behind DPO, we can manipulate the sequences used for gradient updates according to external pragmatic information

$$\begin{aligned} \nabla_{\theta} \mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = \\ -\beta \mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\underbrace{\sigma(\hat{r}_{\theta}(x, y_l) - \hat{r}_{\theta}(x, y_w))}_{\text{higher weight when reward estimate is wrong}} \left[\underbrace{\nabla_{\theta} \log \pi(y_w | x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_{\theta} \log \pi(y_l | x)}_{\text{decrease likelihood of } y_l} \right] \right], \end{aligned}$$

The standard formulation of DPO doesn't help but...

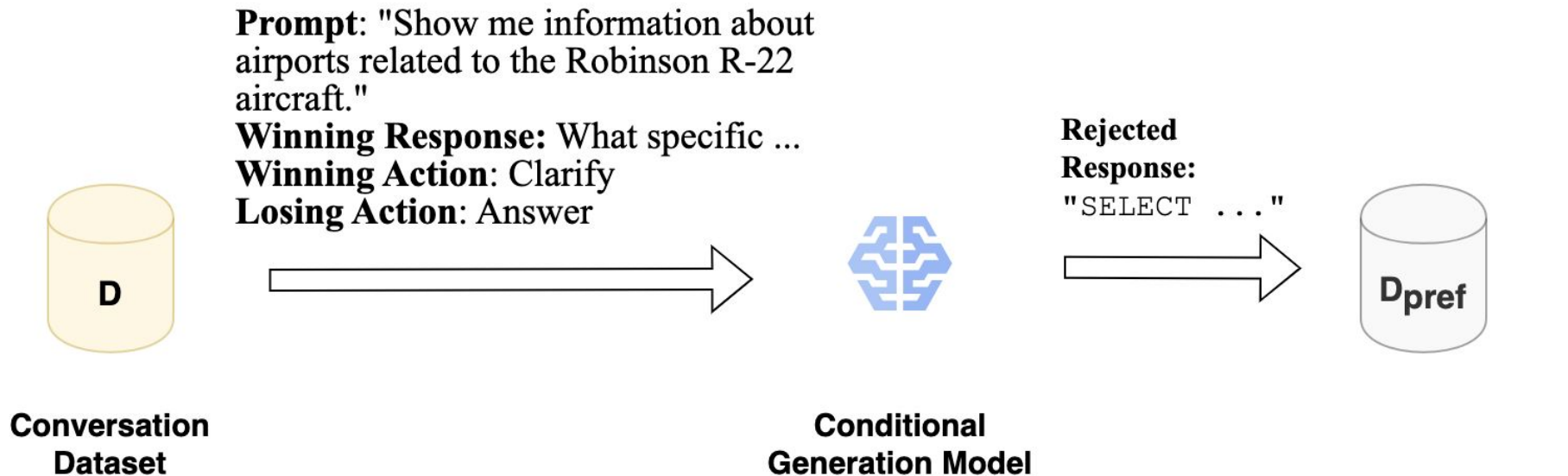
- Contrastive pairings make sense for our setting
- DPO fundamentally relies on log probabilities of token sequences
- Using the intuition behind DPO, we can manipulate the sequences used for gradient updates according to external pragmatic information

We propose ACT: Action-Based Contrastive Self-Training

$$\begin{aligned} \nabla_{\theta} \mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = \\ -\beta \mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\underbrace{\sigma(\hat{r}_{\theta}(x, y_l) - \hat{r}_{\theta}(x, y_w))}_{\text{higher weight when reward estimate is wrong}} \left[\underbrace{\nabla_{\theta} \log \pi(y_w | x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_{\theta} \log \pi(y_l | x)}_{\text{decrease likelihood of } y_l} \right] \right], \end{aligned}$$

Contrastive pairings make sense for our setting

Action Space: {CLARIFY, ANSWER}



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On-Policy Response Sampling



Prompt: "Show me information about airports related to the Robinson R-22 aircraft."

Winning Action: Clarify

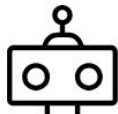
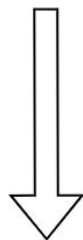
Winning Response: What specific ...

Losing Response:

SELECT avg... FROM ...

Losing Action: Answer

Information Goal: SELECT avg...



[sampled
response y_i]

Policy at Step i

- DPO relies on log probabilities of token sequences

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On-Policy Response Sampling



Prompt: "Show me information about airports related to the Robinson R-22 aircraft."

Winning Action: Clarify

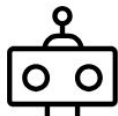
Winning Response: What specific ...

Losing Response:

SELECT avg... FROM ...

Losing Action: Answer

Information Goal: SELECT avg...



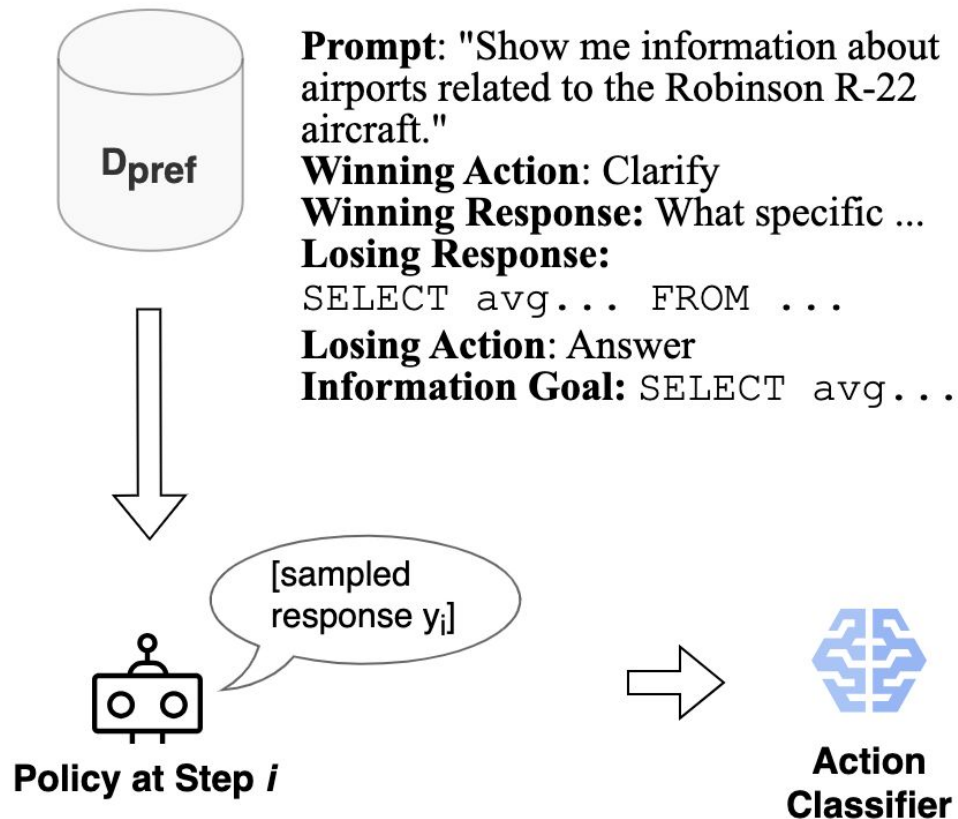
[sampled
response y_i]

Policy at Step i

- DPO relies on log probabilities of token sequences
- A sampled response will likely have a higher log probability than any response from a static dataset or an outside LLM

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On-Policy Response Sampling



Scenario A: Wrong Implicit Action

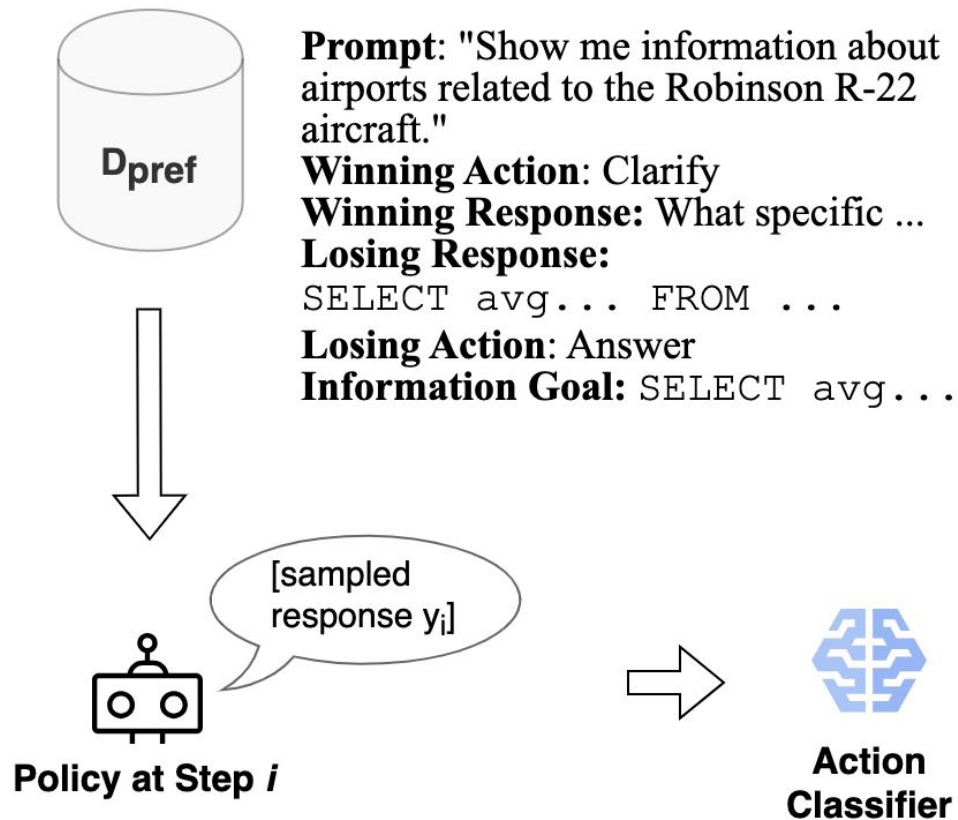
Sampled Response: SELECT ... FROM ...

Detected Action: ANSWER

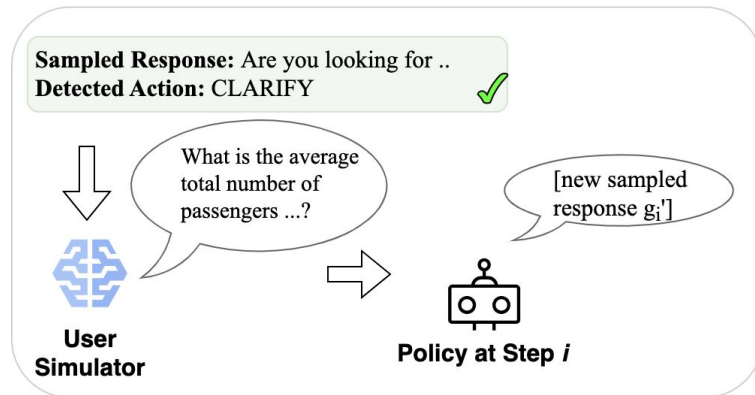


Replace Losing Response with Sampled Response

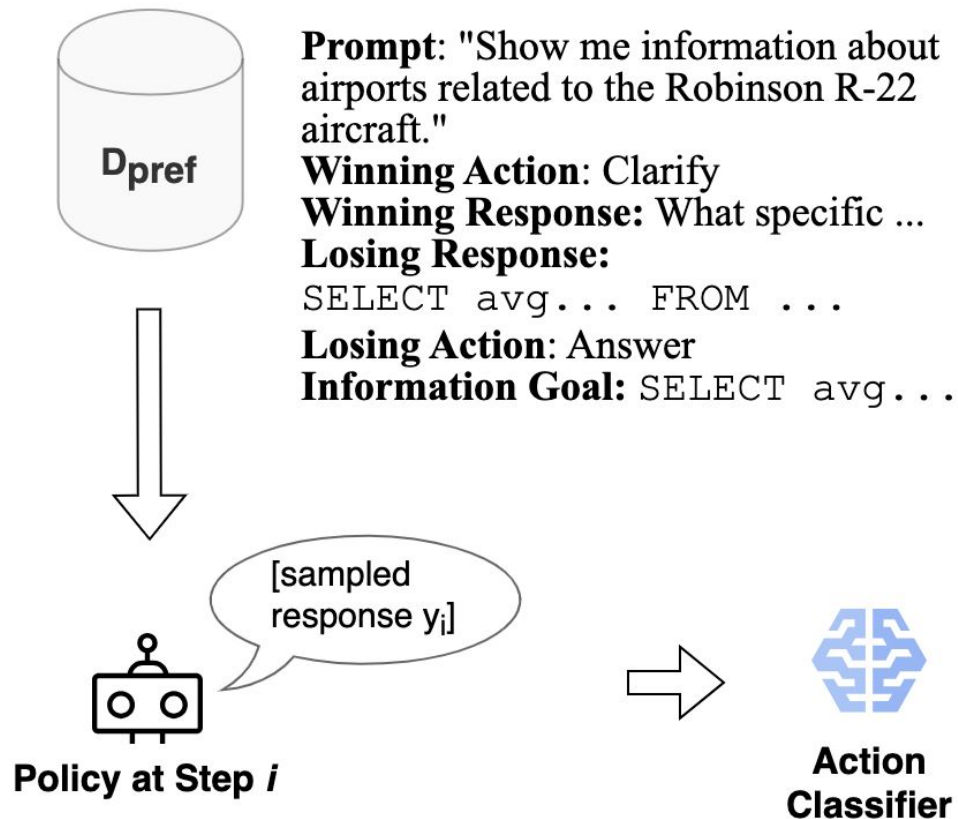
On-Policy Response Sampling



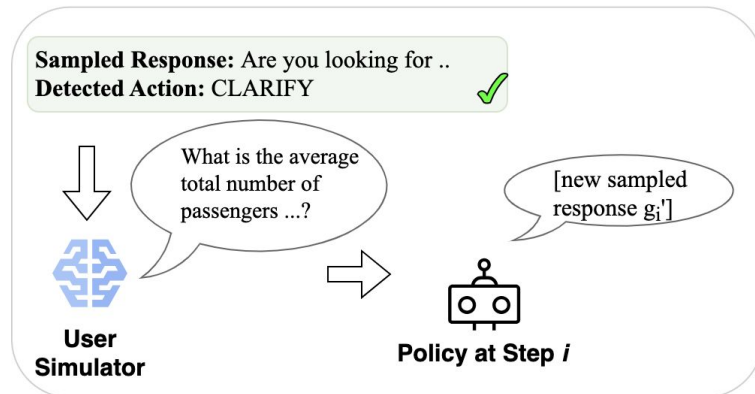
Scenario B: Correct Implicit Action



On-Policy Response Sampling



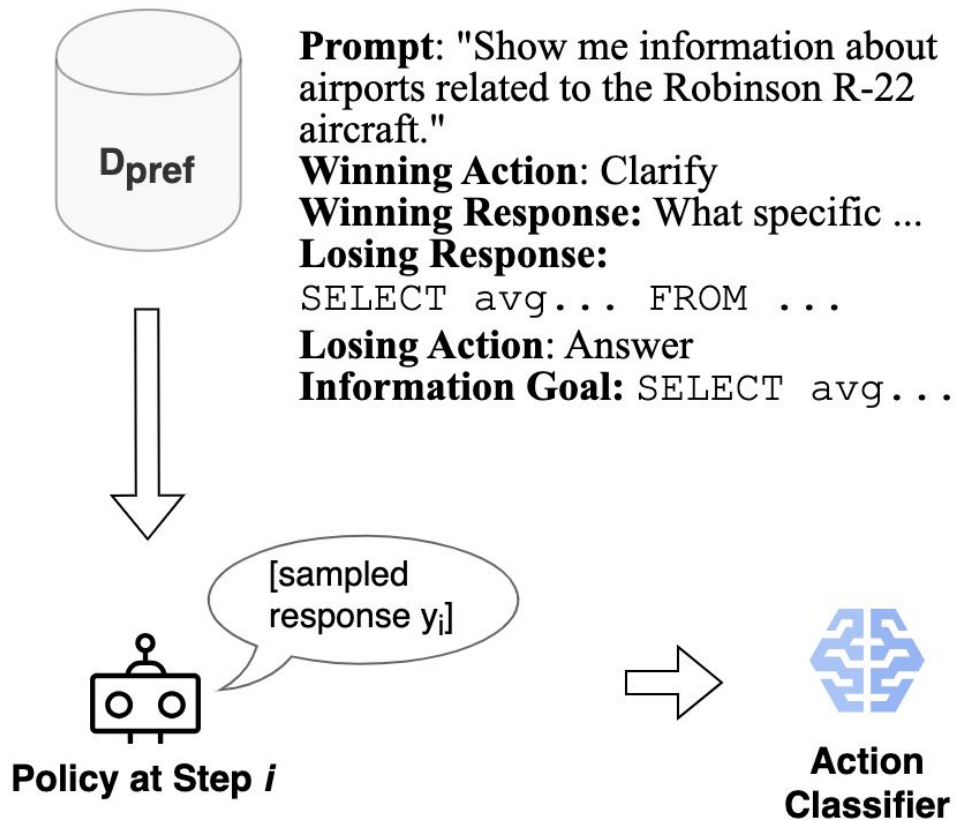
Scenario B: Correct Implicit Action



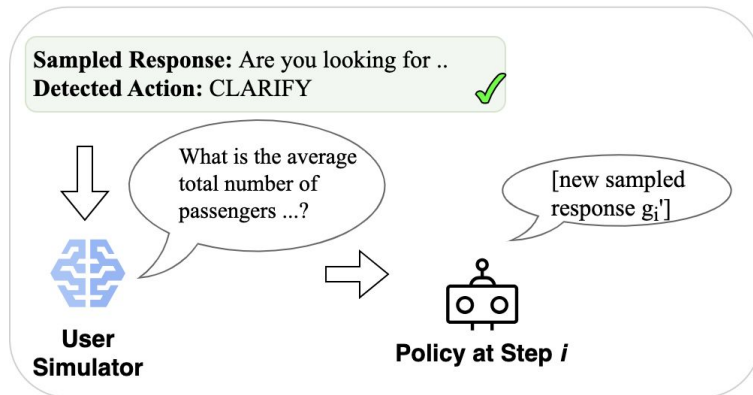
Scenario B1: Incorrect Simulated Outcome



On-Policy Response Sampling



Scenario B: Correct Implicit Action



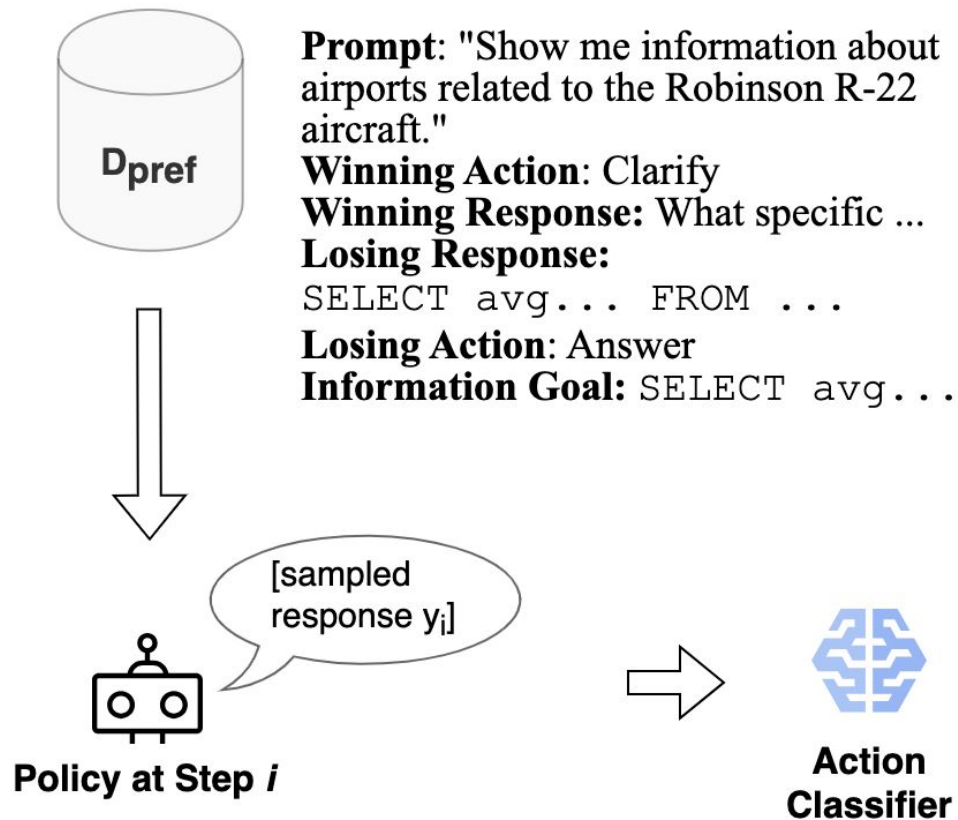
Scenario B1: Incorrect Simulated Outcome



Scenario B2: Correct Simulated Outcome



On-Policy Response Sampling



PACIFIC: Tabular CQA

- Original Metric: single-turn DROP F1 score
- Additional proposed *conversational* evaluations:
 - Action correctness: is the implicit action of an agent's response correct?
 - Agent Response Content correctness: is the agent conversationally skilled enough to arrived at the correct answer through *multi-turn interaction*?

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PACIFIC: Tabular CQA

Adaption Setting			Action-level	Content-level		
Base Model	Approach	Conversations	Macro F1 ↑	Turn F1 ↑	Traj. F1 ↑	Post-Clarify F1 ↑
Gemini Pro	Standard Prompt	10	81.4	59.7	58.7	49.7
Gemini Pro	Chain-of-Thought	10	86.3	66.3	17.1	19.2
Gemini Pro	Proactive MIPrompt	10	78.9	63.4	61.1	18.9
Claude Sonnet	Standard Prompt	10	71.9	43.7	42.0	28.5
Claude Sonnet	Chain-of-Thought	10	80.0	37.2	13.0	6.8
Claude Sonnet	Proactive MIPrompt	10	74.9	47.2	45.9	7.6
Gemini Pro	SFT	50	70.6	46.9	45.6	6.7
Zephyr 7B- β	SFT	50	69.0	57.8	61.3	43.5
Zephyr 7B- β	DPO-Dist (Pro v. Flash)	50	75.5	61.7	55.7	30.8
Zephyr 7B- β	DPO-Dist (Sonnet v. Haiku)	50	74.8	62.0	56.3	31.9
Zephyr 7B- β	<i>ACT</i> (ours)	50	82.2	62.8	61.9	57.2
Gemini Pro	SFT	100	70.3	48.3	46.7	8.3
Zephyr 7B- β	SFT	100	82.3	58.6	60.3	49.9
Zephyr 7B- β	DPO-Dist (Pro v. Flash)	100	68.8	53.3	53.3	31.7
Zephyr 7B- β	DPO-Dist (Sonnet v. Haiku)	100	83.0	59.0	53.7	29.3
Zephyr 7B- β	<i>ACT</i> (ours)	100	86.0	65.0	62.0	57.4
Gemini Pro	SFT	250	88.0	67.4	59.3	10.2
Zephyr 7B- β	SFT	250	86.9	65.1	63.3	56.7
Zephyr 7B- β	DPO-Dist (Pro v. Flash)	250	65.6	53.6	54.1	30.9
Zephyr 7B- β	DPO-Dist (Sonnet v. Haiku)	250	82.8	43.3	38.6	19.6
Zephyr 7B- β	<i>ACT</i> (ours)	250	89.6	68.1	65.7	62.0

Abg-CoQA: CQA for Machine Reading Comprehension

Base Model	Adaptation Setting		Action-level	Content-level	
	Approach	Conversations	Macro F1 $\uparrow$	Turn Similarity $\uparrow$	Traj. Similarity $\uparrow$
Gemini Pro	Standard Prompt	10	55.5	67.0	72.2
Gemini Pro	Chain-of-Thought	10	61.2	63.4	39.1
Gemini Pro	Proactive MIPrompt	10	55.5	63.3	33.3
Claude Sonnet	Standard Prompt	10	66.0	50.1	54.3
Claude Sonnet	Chain-of-Thought	10	63.7	46.2	36.8
Claude Sonnet	Proactive MIPrompt	10	57.2	60.8	32.9
Zephyr 7B- β	SFT	50	44.6	53.3	64.2
Zephyr 7B- β	DPO-Dist (Pro v. Flash)	50	46.9	57.2	61.2
Zephyr 7B- β	DPO-Dist (Sonnet v. Haiku)	50	44.7	57.9	61.5
Zephyr 7B- β	ACT (ours)	50	52.3	66.2	68.8
Zephyr 7B- β	SFT	100	52.6	63.1	69.4
Zephyr 7B- β	DPO-Dist (Pro v. Flash)	100	47.8	61.9	67.1
Zephyr 7B- β	DPO-Dist (Sonnet v. Haiku)	100	44.8	62.0	66.4
Zephyr 7B- β	ACT (ours)	100	51.1	69.5	71.4
Zephyr 7B- β	SFT	250	45.1	61.0	59.8
Zephyr 7B- β	DPO-Dist (Pro v. Flash)	250	46.0	61.9	66.3
Zephyr 7B- β	DPO-Dist (Sonnet v. Haiku)	250	46.3	62.6	67.0
Zephyr 7B- β	ACT (ours)	250	53.3	72.5	75.1

AmbigSQL: Ambiguous Text-to-SQL Generation

- Synthetic dataset synthesized from Spider
- Systematically perturbed every user-request to create ambiguity in one of the following categories:
 - Target columns
 - Target database
 - Presentation format

	Train	Dev	Test
Num. Unambiguous Requests	7,000	1,034	1,034
Num. Ambiguous Requests	7,000	1,034	1,034
Num. Unique Schemas	1,056	145	145
Types of Ambiguity	3	3	3

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AmbigSQL: Ambiguous Text-to-SQL Generation

Model	Ambiguous Request Execution Match	Execution Match with Clarification Turns
Gemini Pro	28.5	68.7
Gemini Ultra	31.2	77.0

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AmbigSQL: Ambiguous Text-to-SQL Generation

Base Model	Adaptation Setting		Action-level	Content-level	
	Approach	Conversations	Accuracy $\uparrow$	Execution Match $\uparrow$	PC Execution Match $\uparrow$
Gemini Pro	Standard Prompt	10	72.1	63.5	75.2
Claude Sonnet	Standard Prompt	10	68.5	66.5	72.4
Zephyr 7B- β	SFT	50	77.4	21.9	13.9
Zephyr 7B- β	DPO-Dist (Pro v. Flash)	50	77.7	42.6	31.5
Zephyr 7B- β	DPO-Dist (Sonnet v. Haiku)	50	78.0	40.9	41.2
Zephyr 7B- β	<i>ACT</i> (ours)	50	80.8	43.6	38.1
Zephyr 7B- β	SFT	100	97.2	43.3	34.3
Zephyr 7B- β	DPO-Dist (Pro v. Flash)	100	98.7	45.1	45.3
Zephyr 7B- β	DPO-Dist (Sonnet v. Haiku)	100	99.8	47.8	44.8
Zephyr 7B- β	<i>ACT</i> (ours)	100	99.2	48.0	49.6
Zephyr 7B- β	SFT	250	99.8	51.0	50.7
Zephyr 7B- β	DPO-Dist (Pro v. Flash)	250	97.3	49.7	44.2
Zephyr 7B- β	DPO-Dist (Sonnet v. Haiku)	250	99.7	50.7	50.3
Zephyr 7B- β	<i>ACT</i> (ours)	250	99.9	52.3	53.0
Zephyr 7B- β	SFT	14,000 (All)	99.8	63.1	60.4

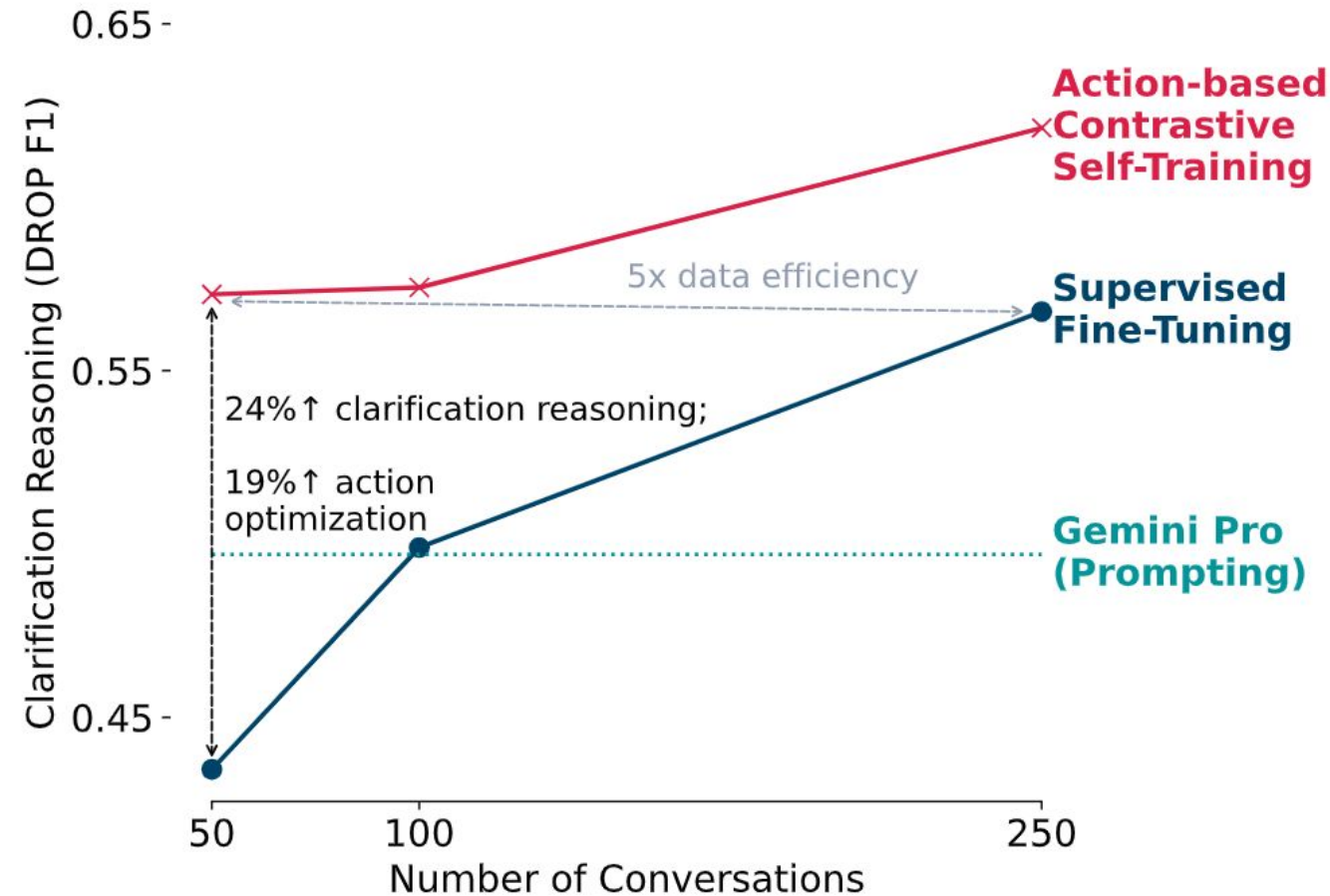
Ablation Studies: 50 PACIFIC Conversations

	Macro F1 ↑	Turn F1 ↑	Traj. F1 ↑	Post-Clarify F1 ↑
Action Importance				
<i>ACT</i> w/ Random Actions	63.2	55.3	58.7	32.8
Ablation of <i>ACT</i> subcomponents				
<i>ACT</i> w/o on-policy sampling	74.8	61.5	59.1	40.5
<i>ACT</i> w/ sampling but w/o simulation	81.4	60.8	60.2	50.1
<i>ACT</i> (full)	82.2	62.8	61.9	57.2
<i>ACT</i> with unaligned foundation models				
Gemma 2B SFT	57.7	38.0	40.5	17.0
Gemma 2B <i>ACT</i>	62.7	42.6	44.0	24.8
Mistral 7B SFT	57.7	53.8	51.4	27.7
Mistral 7B <i>ACT</i>	75.7	58.1	57.6	31.9

Takeaways

- Teaching conversational skills requires conversational objectives
- While SFT can do a lot with abundant data, RL approaches like ACT are more effective in the low data regime
- Conversational model training requires long-horizon optimization

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- Much more data efficient than standard tuning
- Outperforms frontier LLMs even with domain adaptation

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■ Random Actions ■ Offline DPO ■ Online w/o simulation ■ Full ACT

Implicit Action
Recognition



Clarification
Reasoning



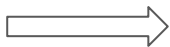
- Each component of ACT greatly contributes to improved performance

0 25 50 75 100

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Modern Challenges in Dialogue Generation

Speaker 1: ...
Speaker 2: ...
Speaker 1: ...



Conversational Agent

[Some policy-optimal dialogue response]

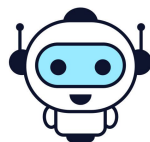
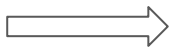
- I. Data Scarcity
- II. Action Policy Optimization
 - A. *Conversation skills require conversational objectives*
 - 1. *Pragmatic Foundations*



"Please turn to page 137 of my new self help book ..."

Modern Challenges in Dialogue Generation

Speaker 1: ...
Speaker 2: ...
Speaker 1: ...



Conversational Agent

[Some policy-optimal dialogue response]

- I. Data Scarcity
- II. Action Policy Optimization
 - A. *Conversation skills require conversational objectives*
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 - 2. *Long-Horizon Optimization*



"Please turn to page 137 of my new self help book ..."